



Scene reconstruction techniques for autonomous driving: a review of 3D Gaussian splatting

Huixin Zhu¹ · Zhili Zhang¹ · Junyang Zhao¹ · Hui Duan¹ · Yao Ding¹ · Xiongwu Xiao² · Junsong Yuan³

Accepted: 10 September 2024 / Published online: 30 November 2024
© The Author(s) 2024

Abstract

As the latest research result of the explicit radiated field technology, 3D Gaussian Splatting (3D GS) replaces the implicit expression represented by Neural Radiated Field (NeRF) and has become the hottest research direction in 3D scene reconstruction. Given the innovative work and vigorous development of 3D GS in autonomous driving, this paper comprehensively reviews and summarizes the existing related research to showcase the evolution of the 3D GS technology and possible future development directions. First, the overall research background of 3D GS is introduced based on two aspects 3D scene reconstruction methods and 3D GS research progress. Second, the relevant knowledge points of 3D GS and the core formulas to clarify the mathematical mechanism of 3D GS are presented. Third, the primary applications of the 3D scene reconstruction technology based on 3D GS in automatic driving are presented through new perspective synthesis, scene understanding, and simultaneous localization and map building (SLAM). Finally, the research frontier directions of 3D GS in autonomous driving are described, including structure optimization, 4D scene reconstruction, and cross-domain research. This paper may provide an effective and convenient pathway for researchers to understand, explore, apply this novel research method, and promote the development and application of 3D GS in automatic driving.

Keywords 3D GS · Explicit radiation field · Autonomous driving · 3D scene reconstruction · Real-time rendering

✉ Zhili Zhang
157918018@qq.com

✉ Junyang Zhao
zhaojy802@sina.com

¹ Laboratory of Intelligent Control, Xi'an Research Institute of High Technology, Xi'an 710025, China

² The State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China

³ The Computer Science and Engineering Department, State University of New York, Buffalo, NY 14228, USA

1 Introduction

Among the 3D scene reconstruction methods, active reconstruction techniques perform poorly the application of outdoor scenes due to their high equipment cost, slow detection speed, and poor algorithm generalization (Zhou et al. 2024a). For example, the structured light method can perform 3D reconstruction with high accuracy, but it is not applicable to outdoor scenes and is often used in industrial inspection and cultural relic restoration (Zheng et al. 2020). To tackle this, with the fast acquisition speed, high equipment efficiency, and high flexibility and robustness of the overall system, the passive reconstruction technique, which focuses on the multi-view stereo vision method (Li et al. 2019), has become a popular research area with critical complexity in outdoor 3D scene reconstruction technology. The core idea of the traditional multi-view stereo vision method is to constrain and optimize the feature points and parameters through geometric principles. Then, the multi-view stereo vision method can accurately recover the three-dimensional structure of the scene from multiple two-dimensional images. The radiation field method emerged with the development of deep learning to address the shortcomings of the traditional methods, such as feature matching and low reconstruction accuracy, and it has shown high performance (Yan et al. 2021). Neural Radiance Field (NeRF) (Mildenhall et al. 2020a), which is a representative of the implicit radiation field in computer vision, is favored by researchers for its highly realistic rendering quality, accessible and flexible scene representation, and end-to-end learning framework. Nonetheless, NeRF models require a large amount of computational resources during training and have a slow rendering speed, which makes them difficult to integrate into in-vehicle systems to fulfill the real-time reliability requirements of high-level autonomous driving (Zhu et al. 2023a). 3D Gaussian Splatting (3D GS) (Kerbl et al. 2023) combines the data storage advantage of the explicit radiation field and network optimization characteristics of the implicit radiation field to achieve high-speed operation, high-quality rendering, and high-reliability interactions. 3D GS refreshes the SOTA of various indicators of the 3D scene reconstruction and expects to provide strong technical support to realize high-level autonomous driving.

In terms of the rendering method, 3D GS abandons the ray tracing method represented by NeRF and uses the fast micronizable rasterization method of a Graphics Processing Unit (GPU) for rendering, which significantly accelerates the training process and can render in real time. 3D GS uses ellipsoid primitives to explicitly represent the entire “body data”. And 3D GS is isomorphic to an ellipsoid, which can be obtained by deflating and rotating the ball axially. Compared with other geometrical shapes (e.g., rectangle), ellipsoid has continuity and microscopcity, thus being able to adapt to the irregular distribution of the point cloud, in addition to having better interpretability, computational efficiency, and robustness. Thus, 3D GS selects the ellipsoid representation as a tiny microscopic space to flexibly and efficiently characterize the data distribution of a realistic 3D scene. Subsequently, 3D GS constructs the point cloud with a covariance while determining the central axis and size of the ellipsoid by analyzing the eigenvalues and eigenvectors of the covariance matrix. 3D GS can be sampled to obtain a similar Gaussian collection to the original data distribution. For the rasterization of the ellipsoid, it is necessary for the researcher to implement it in CUDA, where the process of projecting the ellipsoid onto the projection plane to get a 2D figure is called “Splatting”. In optimizing the Gaussian body, 3D GS densifies the Gaussian body parameters (e.g. 3D coordinates, transparency, anisotropic covariance, and spherical

harmonic function) by interspersing adaptive control steps into the parameter optimization process.

Figure 1 shows the relationship between autonomous driving and 3D reconstruction technology. Autonomous driving requires 3D reconstruction technology to help vehicles better understand surroundings (e.g. driving scenario understanding, simultaneous localization and mapping, and urban scene reconstruction). The reconstruction technology reconstructs the environment around the vehicle, providing more accurate information about the environment for the autonomous driving system. This helps autonomous driving systems to better plan driving paths, avoid collisions, and improve driving safety. The emerging 3D GS technique is an efficient 3D reconstruction method that works by mapping point cloud data from the surrounding environment onto a 3D grid, using a Gaussian distribution function to represent the position and size of each point. This method can represent the 3D information of the surrounding environment more accurately and improve the sensing ability of the autonomous driving system, enabling it to detect and recognize the surrounding obstacles more accurately. In addition, 3D GS can effectively deal with the sparsity and noise problems of the point cloud data and improve the stability and reliability of the reconstruction results.

Current literature has focused on 3D GS but much less in autonomous driving application, which need explore its underlying advancements and applications in autonomous driving. To popularize and promote the 3D GS-based scene reconstruction technology in autonomous driving, this paper compiles the relevant subject materials of 3D GS. The excellent overall performance and vigorous development trend of 3D GS are explored along with the development of the 3D GS-based scene reconstruction technology, its application, current challenges, and research frontiers in autonomous driving. Figure 2 shows that the structural of this survey is organized as follows: Sect. 2 introduces the research background of 3D GS, including 3D scene reconstruction methods and the relevant research progress of 3D GS. Section 3 introduces the main modules of 3D GS and reveals the derivation process of 3D GS core formulas. Section 4 discusses the application of 3D GS in three key aspects

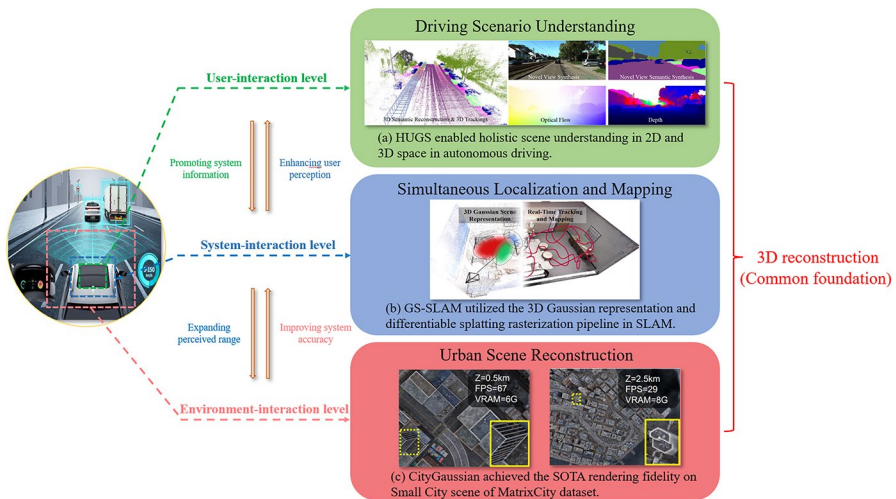


Fig. 1 Schematic diagram of the relationship between autonomous driving and 3D reconstruction technology (Zhou et al. 2023; Yan et al. 2024; Liu et al. 2024)

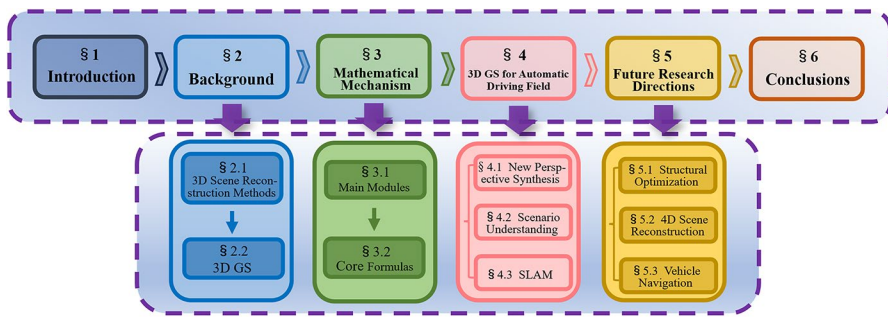


Fig. 2 The framework of this survey

of autonomous driving, that demonstrate its flexibility and efficiency advantages. Section 5 emphasizes further research directions of 3D GS in autonomous driving to explore its potential capabilities. This paper aims to help the readers easily understand the research of 3D GS, which is an emerging technology in autonomous driving. In addition, it aims to help researchers think about how to realize high-level autonomous driving.

2 Background

2.1 3D Scene reconstruction methods

3D scene reconstruction for autonomous driving involves using sensor data to create a 3D model or description of the vehicle's surroundings. The purpose of this step is to provide a real-time, accurate understanding of the environment for the autonomous driving system, which will facilitate autonomous navigation, obstacle avoidance, decision planning, and other driving-related safe tasks. 3D scene reconstruction methods can be categorized into active and passive reconstruction techniques based on how the device collects data.

Compared with active reconstruction methods, passive reconstruction methods can obtain image data based on natural scene illumination and solve the three-dimensional spatial information of objects through specific algorithms. Because of its advantages of low equipment cost, fast deployment, and high robustness of algorithms, passive reconstruction methods are widely used in outdoor scene reconstruction. Representative methods for automatic driving are Monocular Stereo Vision (MSV) (Jiang et al. 2022), Binocular Stereo Vision (BSV) (Zhai et al. 2020), and Multi-View Stereo (MVS) (Wang et al. 2021a). MSV can reconstruct a 3D scene by using only one camera device to take photos, and its representative algorithms are Shape from Shading (SFS) (Bao et al. 2023), Shape from Texture (SFT) (Lobay and Forsyth 2006), and Shape from Silhouettes/Contours (SFS/SFC) (Phothong et al. 2018; Fan et al. 2022). Since conventional monocular vision systems cannot directly measure the depth, they must rely on visual cues and a priori knowledge to estimate the depth, which increases the complexity of the algorithm and reduces the accuracy and real-time performance of the system. With the popularity of consumer-grade RGB-D cameras (e.g., Microsoft's Kinect series, Intel's RealSense D400 series, Orbbec's Astra series), several studies have been conducted to reconstruct a scene in 3D based on directly acquired 3D coordinate information. The representative algorithms are Dynamic Fusion (Newcombe et

al. 2015), and Bundle Fusion (Dai et al. 2017). Binocular cameras simulate the human visual system by generating a parallax in the design of the camera mounting position, then obtaining the depth information of objects. The main idea behind BSV is to calibrate the internal and external parameters of the camera, and subsequently match the features under Euclidean geometry based on the idea of antipodal geometry. Then, BSV uses the principle of triangulation to obtain the depth information and realize the three-dimensional reconstruction. The traditional feature extraction algorithms are HOG (Shirpour et al. 2023), SIFT (Somaya et al. 2018), ORB (Song et al. 2018), and LBP (Wang et al. 2022). With the explosion of the extended-depth CNN model AlexNet (Nebauer 1998) in 2012, deep learning methods have been making significant breakthroughs in 3D reconstruction and attracting the attention of researchers worldwide. To be concrete, binocular depth estimation algorithms based on deep learning mainly include GC Net (Kendall et al. 2017), PSMNet (Chang and Chen 2018), and StereoDRNet (Chabra et al. 2019). However, the difficulty of BSV in practical applications lies in the feature matching stage, where the matching accuracy of the left and right camera images significantly affects the subsequent imaging effect of the algorithm.

MVS increases the number of cameras and viewpoints based on BSV. Moreover, MVS establishes a dense correspondence based on the association of multiple images with known camera poses, to obtain a dense 3D point cloud on the surface of the scene objects (Furukawa and Hernandez 2013). MVS does not depend on specific external environmental conditions. With the advantages of high accuracy, fast acquisition, and low cost, MVS is a popular research field and a key area of difficulty in 3D scene reconstruction. Based on different processing methods, MVS can be classified into two categories: non-deep learning and deep learning-based frameworks. And the main mathematical representations of the scene geometry are voxels, meshes, point clouds, and depth maps (Mustafa et al. 2021). Non-deep learning-based MVS reconstruction methods rely on the ideas of parallax and geometric constraints to reconstruct 3D scene. Non-deep learning-based methods usually use more mature computer vision algorithms, such as feature extraction, matching, and optimization. Thus, non-deep learning-based methods, which can be applied in a wide range of scenes, highly depend on texture-rich regions, perform poorly on texture-poor or repetitively textured regions, and face significant challenges in complex scenes such as those with occlusions, reflections, and transparent objects. Deep learning-based methods introduce excellent ideas from the neural network and improve the reconstruction quality, automation, and efficiency. MVSNet (Yao et al. 2018) extends the depth estimation method of binocular stereo matching with the cost volume of two images to a multi-view depth estimation and proposes a differentiable univariate responsive transform based on the cost volume. Based on MVSNet, subsequent improvement works [e.g. R-MVSNet (Yao et al. 2019), Fast-MVSNet (Yu and Gao 2020), and EPP-MVSNet (Ma et al. 2021)] have mainly improved the framework structure, model size, and computational speed, and successfully applied depth information to the MVS problem. Deep learning-based MVS reconstruction methods can fully consider the ambient illumination changes, encode the scene information, and extract local features, which significantly facilitates the 3D reconstruction of non-Lambertian surfaces and weakly textured regions (Zhang et al. 2023b). How to further develop an end-to-end MVS system that integrates all reconstruction steps (including depth estimation, matching, filtering, and fusion) into a unified framework for better performance and simpler pipeline is the next key direction of research.

NeRF technology has attracted much attention from researchers since its introduction in 2020 (Mildenhall et al. 2020a). NeRF-based 3D scene reconstruction algorithms have excellent multi-view consistency and continuity, do not require explicit 3D annotation, and are adaptable and easily scalable. Therefore NeRF-based algorithms have become some of the most essential research methods for scene reconstruction. NeRF introduces two breakthroughs in the optimization of Multilayer Perceptron (MLP): Positional Encoding (PEC) and Hierarchical Volume Sampling (HVS). PEC helps enhance the network's ability to accurately characterize the high-frequency details of an image, whereas HVS significantly improves the image rendering efficiency. As shown in Fig. 3, the scene representation process of the benchmark NeRF consists of four main parts (Zhang et al. 2023a): input of 5D vectors, output of the color and voxel density information, voxel rendering to synthesize the pixel colors, and a loss function to optimize the MLP parameters. The positional coding technique can map the input 5D vectors to high frequencies and facilitate the learning process of the neural network. The multilayer voxel sampling technique can simultaneously optimize the coarse and fine networks to obtain more sampling points without densely sampling all regions on the ray. However, the NeRF method has a computational cost and high data demand during training and rendering. It has limited processing capability for dynamic scenes and real-time applications, so further development and improvement are necessary (Kim et al. 2023; Zhang et al. 2023b; Yang et al. 2023a).

2.2 3D GS

Despite improvements in various aspects, NeRF mainly uses implicit coordinate-based models to map spatial coordinates to pixel values and directly renders using body rendering and neural networks, which are computationally demanding, take too long to render, and are costly to train. In this context, the 3D Gaussian Splatting (3D GS) (Kerbl et al. 2023) technique emerged as a paradigm-shift approach to redefine the boundary between scene reconstruction and rendering (Chen and Wang 2024). As shown in Fig. 4, compared with the mainstream NeRF algorithm, 3D GS retains the advantages of high-quality scene reconstruction and ensures real-time rendering effects at the SOTA level in a short time period. Hence, 3D GS became an essential and prominent research methodology across the field of 3D scene reconstruction by the end of 2023.

In 3D reconstruction, recovering a 3D scene from a sequence of images is one of the most fundamental problems. Several researchers have noted that ellipses (for which circles are a particular case) are among the most commonly used features for 3D position esti-

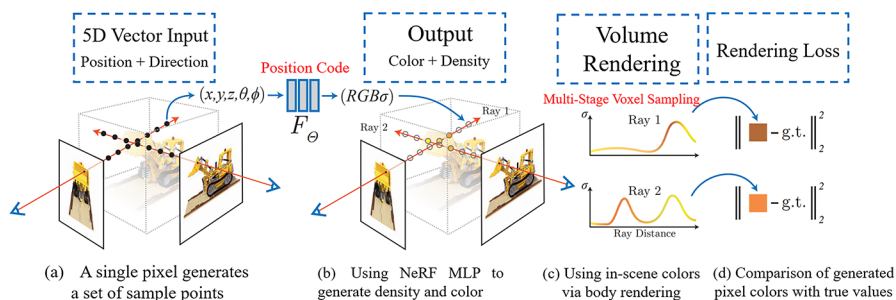


Fig. 3 Scene representation flow for the benchmark NeRF (Mildenhall et al. 2020a)

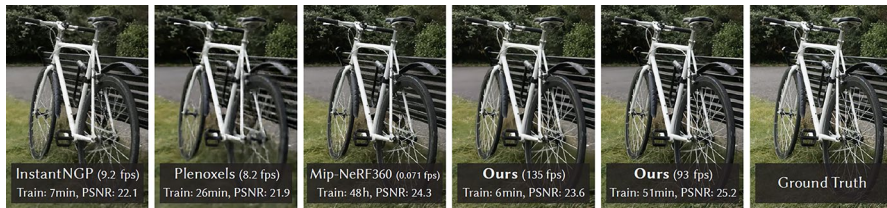


Fig. 4 Comparison of the reconstruction quality and runtime speed between 3D GS and mainstream NeRF algorithms (Kerbl et al. 2023)

mation. Ellipses have a higher image localization accuracy than points or line segments (Mcvey and Jarvis 1997; Young 1986). Zwicker et al. (2002a) proposed a high-quality splatting method based on an elliptic Gaussian kernel, called EWA Splatting. The article stated that EWA Splatting operated with an elliptical Gaussian kernel, which provided high-quality images without aliasing artifacts or excessive blurring of the body data and could efficiently handle aspherical volume cells. Elliptic Gaussian body and sputtering methods apply to regular, linear, and irregular volume datasets, so they demonstrate great adaptability and convenience, and have attracted extensive attention from researchers. Wang et al. (2005) constructed a sputtering model based on an orientation reconstruction filter kernel to address the problem of unclear object edges in volume rendering. Due to the shortcomings of the traditional spherical Gaussian function in 3D scenes (particularly when object surface orientation or texture information is involved), Wang et al. (2005) used a directional, continuous elliptical Gaussian footprint function to compress the erroneous gradient estimation in projection, which yielded finer details of the 3D scene.

In the process of using conic curves as basic image features for research (Long 1996), the algorithms based on ellipses for attitude estimation, stereo construction and motion estimation have proved their excellent performance. However, the prior information of the camera, number of images, and geometrical relations constrain the application and development of ellipsoid structures in scene reconstruction. Different researchers have proposed many different solutions. Mai et al. (2010) proposed a method to reconstruct 3D ellipses (including circles) from a sequence of 2D images taken by an uncalibrated camera. The minimization problem in the transformation was transformed into a series of minimization subproblems, which enabled the simultaneous reconstruction of multiple 3D ellipses from multiple images and can easily handle missing ellipses and partially occluded images. Garcia et al. (2015) approximated the 3D point cloud data to a basic elliptic model in scene reconstruction. The model was also rendered as a set of textured elliptical splashes using multi-texturing. The results showed that their system dramatically improved the photorealism of the rendered point cloud while significantly reducing the modeling size. Other articles provide specific implications regarding mathematical mechanisms and model architecture (Ivo et al. 2020; Anjos et al. 2018). For example, Wang et al. (2019) proposed a high-fidelity differentiable point cloud renderer algorithm, which was called the Differentiable Surface Splatting (DSS) method. DSS handles the gradient of the point positions and normal after reasonable settings. In addition, it is beneficial to enhance the ability of DSS by handles the discontinuities accurately and robustly in the rendering function, large-scale topological changes and small-scale detail modifications without explicit connections.

Since the 3D GS project was open-sourced by the Université Côte d'Azur (UAC) in France in August 2023 (Kerbl et al. 2023), the academic community has been, extremely active with various emerging derivative models. In autonomous driving, 3D GS has similarly revolutionized 3D scene reconstruction and local environment perception techniques. Based on the existing open research, Table 1 shows the relevant datasets for to evaluate the performance of 3D GS in autonomous driving.

3 Mathematical mechanism

The interpretability of 3D GS lies in its explicit structural expression and data storage capability. An explanation of the mathematical mechanism, i.e., the working principle, of 3D GS can help researchers analyze the complex relationships in 3D scene reconstruction technology and reveal the performance characteristics of 3D GS in depth. In this chapter, we will start from the main modules of 3D GS to introduce the overall computing process of 3D GS and focus on the mathematical derivation of the core formula.

3.1 Main modules

A significant difference between NeRF and 3D GS is the explicit and implicit representations of the radiation field. The radiation field is a model that quantifies the intensity, distribution, effect, and other parameters of light in three-dimensional space and can be represented by Eq. (1).

$$F(\mathbf{x}, \theta, \phi) \rightarrow (\mathbf{c}, \sigma) \quad (1)$$

where $\mathbf{x} = (x, y, z)$ represents the 3D scene coordinates, (θ, ϕ) represents the azimuthal and polar angles in spherical coordinates, $\mathbf{c} = (r, g, b)$ represents the color, and σ is the bulk density. In the field of autopilot, the radiation field can reconstruct 3D scenes and perform rendering in real time by deconstructing the interaction between light and the surface material of the environmental target. The strategy of NeRF to reconstruct a 3D scene is to not directly use the explicit scene set shape to represent the light distribution of the scene and to directly synthesize the image in the new perspective only based on the positional internal reference and the image. First, the 3D Cartesian unit vector $\mathbf{d} = (d_x, d_y, d_z)$ is to describe (θ, ϕ) ; this process is encoded with sine and cosine transformations. So, this process can increase the frequency component of the input and help the neural network learn more complex functions. NeRF uses a Multilayer Perceptron (MLP) as its neural network structure to accomplish the output of Eq. (1) (Mildenhall et al. 2020a), where the output is $(\mathbf{c}, \sigma)$ at each location. Then, NeRF calculates the final pixel color by sampling multiple points along the camera rays, using a neural network to predict the $(\mathbf{c}, \sigma)$ values of these points, and accumulating the color and transparency along the rays via integration. Ultimately, the reconstructed and authentic images are compared by a gradient descent to optimize the neural network weights and speed up the operation.

Compared to mesh- or voxel-based methods, the rendering performance of NeRF-recorded scenes is much slower if MLP is exclusively used. Based on specific application requirements, available computational resources, and the trade-off between rendering qual-

Table 1 Relevant datasets to evaluate the performance of 3D GS in autonomous driving

Dataset	Time	Areas of focus	Description	Related 3D GS Algorithms
Mip-NeRF 360 (Barron et al. 2022)	CVPR 2022	Scene Renderer	An extended version of Mip-NeRF that consists of 9 scenes with 5 outdoors and 4 indoors, each of which contains a complex central object or area and a detailed background	3D GS (Kerbl et al. 2023)
Blender (Mildenhall et al. 2020b)	ECCV 2020	Scene Renderer	A widely used synthetic dataset generated from 3D models and scene renderings created by the Blender software. This synthetic training set can provide the ideal training environment to help models learn to recognize basic features without noise or bias	3D GS (Kerbl et al. 2023), Mip-Splatting (Yu et al. 2023)
CO3D-V2 (Reizenstein et al. 2021)	CVPR 2021	Scene reconstruction	A smaller, downloadable single-sequence subset of approximately 100 sequences, which include only the sequences used to evaluate the multi-view single-sequence task. All images were cropped to the same height×width	CF-3DGS (Fu et al. 2023)
Deblur-NeRF (Ma et al. 2022)	CVPR 2022	Scene reconstruction	A dataset of 20 natural scenes, with 10 scenes of each blur type. Camera motion blur images were captured using a Canon EOS RP in the manual exposure mode, whereas reference images were taken using a tripod	Deblurring 3D-GS (Lee et al. 2024)
Replica (Straub et al. 2019)	2019	RGB-D SLAM	A dataset that includes challenging pure rotational camera motions and is designed for the RGB-D SLAM evaluation	Gaussian Splatting SLAM (Hidenobu et al. 2023), GS-SLAM (Yan et al. 2023), SplatAM (Keetha et al. 2023)
HyperNerf (Park et al. 2021)	2021	Indoor SLAM	A dataset that focuses on topological changes, including indoor life-like scenarios such as human eyes opening and closing, peeling banana peels, 3D printed chick toys, and broomstick morphing	Deformable 3D Gaussians (Yang et al. 2023b)
Scan-Net++ (Yeshwanth et al. 2023)	ICCV 2023	Indoor SLAM	A dataset that includes high-quality color and geometry capture data, normal-grade RGB-D videos, 33-megapixel images and RGB-D streaming data. Each scene was captured at sub-millimeter resolution using a high-end laser scanner	SplaTAM (Keetha et al. 2023)
TUM RGB-D (Sturm et al. 2012)	IROS 2012	Indoor SLAM	A dataset of data recorded based on the total frame rate (30 Hz) and sensor resolution (640×480 pixels). Ground-truth trajectories were obtained by a high-precision motion capture system with eight high-speed tracking cameras (100 Hz)	Gaussian Splatting SLAM (Hidenobu et al. 2023), GS-SLAM (Yan et al. 2023), SplatAM (Keetha et al. 2023)
the Tanks and Temples (Knapitsch et al. 2017)	2017	Outdoor reconstruction	A large dataset that is widely used in the MVS field. It can verify that the model can more accurately reconstruct scenes with large changes in illumination and the presence of dynamic targets	3D GS (Kerbl et al. 2023), CF-3DGS (Fu et al. 2023), Compact-3D GR (Lee et al. 2023)

Table 1 (continued)

Dataset	Time	Areas of focus	Description	Related 3D GS Algorithms
FusionPortable (Jiao et al. 2022)	IROS 2022	Multimodal fusion	A dataset that supports odometry, localization, mapping, and some sensing tasks, to provide the ground truth for decoupled localization and mapping performance evaluation	LIV-GaussMap (Hong et al. 2024)
FASTLIVO (Zheng et al. 2022)	IROS 2022	Multimodal fusion	A dataset that focuses on providing non-repetitive LiDAR data in an open environment	LIV-GaussMap (Hong et al. 2024)
DNeRF (Albert et al. 2021)	CVPR 2021	Synthetic datasets	A dataset of scene data for monocular dynamic scenes. The camera poses were near-randomly generated for each timestamp. Each scene in the dataset contains 50–200 dynamic frames	Deformable 3D Gaussians (Yang et al. 2023b), 4D-GS (Wu et al. 2023)
Gamba (Shen et al. 2024)	2024	Scene reconstruction	A dataset that concludes a final training set of approximately 20 k high-quality 3D objects	Gamba (Shen et al. 2024)
PhysGaussian (Xie et al. 2023)	CVPR 2024	Scene reconstruction	A dataset that concludes two real-world datasets (referred to as toast and jam) with an iPhone. Each scene contains 150 photos. The initial point clouds and camera parameters are acquired by COLMAP	PhysGaussian (Xie et al. 2023)

ity and speed, many researchers have gradually adopted explicit methods to improve NeRF to obtain fast, accurate and generalizable algorithmic models (Zhang et al. 2023c). Nvidia's Mueller et al. (2022) combined explicit and implicit approaches to propose the Instant NGP model, which excludes the efficiency problem of the traditional NeRF model in parameterizing fully connected neural networks. In Instant NGP, the explicit module maps significant voxels to features. Thus, the implicit module can perform the MLP learning through the design and implementation of multi-resolution hash coding, which significantly improves the rendering speed without losing accuracy. Fridovich et al. (2022) proposed a new explicit model, called Plenoxels. Plenoxels can represent a scene using a set of sparse, learnable, spherical harmonic functions of the 3D mesh, where these coefficients define a continuous bulk density and color function. The results show that the rendering speed of explicit Plenoxels greatly improves, but the computational overhead inevitably increases with the introduction of voxels.

Explicit radial field representations such as voxels and point clouds should be used for 3D scene reconstruction techniques. Because they are well suited for fast GPU/CUDA-based rasterization. For complex or unstructured 3D shapes, explicit radial field representations such as voxels and point clouds can also directly simulate the interaction of light with the scene, provide accurate geometric information, and facilitate multi-scale scene representation from macro to micro (Chen et al. 2023a). 3D GS adopts a flexible and efficient representation strategy to reconstruct a 3D scene using 3D Gaussian and reasonably optimizes the network characteristics of NeRF by combining the series parameters. This fusion of explicit and implicit scene reconstruction models takes advantage of the query-able and editable nature of explicitly stored data while absorbing the automaticity and accuracy of implicitly matched optimized data, ensuring efficient shape representation, and high-quality rendering capability, and providing fast training speed and real-time performance.

The original 3D GS model based on three main modules: 3D Gaussian representation, attribute optimization, and real-time rendering. Figure 5 shows its main flowchart. In the 3D Gaussian representation module, the camera used in 3D GS is calibrated by the Structure from Motion (SfM), and high-quality results can be obtained by only inputting the SfM sparse point cloud when initializing the 3D Gaussians set. This strategy follows the NeRF input strategy but outperforms most point-based solutions that require MVS data. 3D GS also performs well on randomly initialized NeRF synthetic datasets. The superiority is that 3D Gaussian as volumetric representations can be micro-supplied for smooth rendering and projected into a 2D space for efficient rasterization using α -blending (alpha-blending). In the attribute optimization module, various attributes of 3D Gaussian (including the Gaussian body mean and covariance in the world coordinate system, ellipsoid position, shape, opacity, rotation matrix, magnitude, anisotropic covariance, and spherical harmonic (SH) coefficients) must be tuned and optimized to acceptably control the behavior of the Gaussian function. In practice, optimization strategies such as gradient descent, backpropagation, and regularization, can be used in conjunction with an adaptive density control step to add, replace, and remove 3D Gaussian. In the real-time rendering module, 3D GS draws on tile-based rasterization research (Christoph and Michael 2021) to achieve fast GPU sorting. 3D GS divides the spatial 3D scene into a series of tiny tiles. Then, each tile performs the local sorting of the Gaussian function, utilizes the parallel processing capability of the GPU to simultaneously perform reasonable global sorting of the data in multiple tiles, and optimizes the memory access pattern to reduce the use of memory bandwidth to improve the sorting efficiency. Meanwhile, the Anisotropic Splatting technique of 3D GS to render the 3D Gaussian function considers the surface orientation and texture of the Gaussian body, so it can produce a more realistic effect in the rendering process. 3D GS uses sorting and α -blending to normalize the order of visibility of the Gaussian and track the required traversal of the sorted splats to significantly reduce the backpropagation time-consuming. Thus, it achieves real-time rendering in SOTA quality at high resolution. Above all, 3D GS successfully solves the trade-off between speed and quality of existing algorithms in the 3D reconstruction domain by tightly integrating three key modules. The 3D Gaussian representation module provides a compact and continuous description of complex 3D shapes, the attribute optimization module ensures that the Gaussian function can accurately match the input data, and the real-time rendering module leverages the efficiency of GPU-accelerated algorithms to achieve high-quality rendering, fast rendering, and dynamic updates. These three modules complement one another and together achieve the ideal balance between

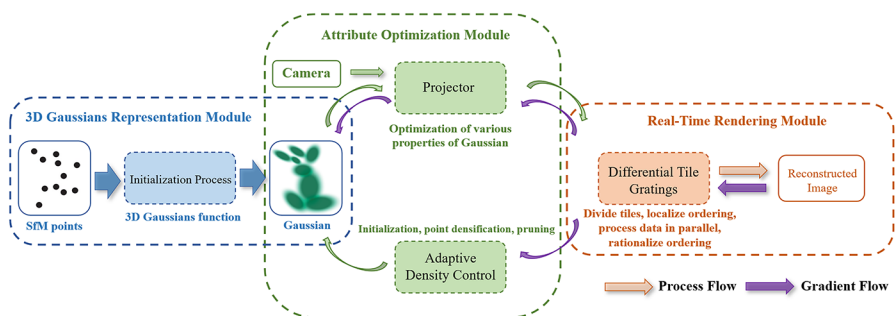


Fig. 5 Schematic flow diagram of the three major modules of 3D Gaussian Splatting (Zhu et al. 2023a)

speed and quality, given that 3D GS can provide fast response and real-time performance while maintaining high visual quality (He et al. 2024).

3.2 Core formulas

The differentiable 3D GS method has the properties of differentiable volume representation, which can back-propagate the gradient of the rendering results to achieve gradient-based optimization. Instead of dividing the 3D space into regular grids as in the traditional structured voxel representation, the 3D space can be directly sampled and rendered at arbitrary positions. By sampling the Gaussian distributions of the points in the 3D space, and calculating the color and density based on the sampling results, the rendering process can be clarified, so fast hybrid rendering is achieved. 3D GS can explicitly render the color and density, which enables fast hybrid rendering. Based on the above properties, differentiable 3D GS accomplishes the goal of the high-quality new view synthesis based on undirected sparse point sets and can be used as an efficient scene representation for 3D scene reconstruction. The mathematical derivation part of 3D GS consists of three parts: representation of multivariate Gaussian functions, initialization of Gaussian bodies, and splatting of Gaussian bodies (Kerbl et al. 2023; Zwicker et al. 2001, 2002b).

3.2.1 Representation of multivariate Gaussian functions

In 3D GS, Kerbl et al. (2023) followed the Gaussian representation function of Zwicker et al. (2001) in EWA Splatting. Equation (1) shows that their multivariate Gaussian function is defined under world space using a 3D covariance matrix Σ centered at point μ . μ and Σ denote the 3D Gaussian position and anisotropic covariance corresponding to the 3D Gaussian position and anisotropy, respectively. Compared to the standard form, the 3D GS removes the scale factor in front of the exponential part (but this does not affect the ellipsoid geometry), and the default model coordinates are centered at the origin, which is convenient for rotating and deflating, and then adding translations when putting into world space. The function is multiplied by the coefficient α in the subsequent mixing process to avoid the problem of average estimation caused by the extreme sparsity of SfM points.

$$G(x) = e^{-\frac{(x)^T(\Sigma)^{-1}(x)}{2}} \quad (2)$$

3.2.2 Initialization of Gaussian

About 80% of the central part of 3D GS uses the PyTorch deep learning framework to build the code operations, which leads to some architectures needing to be adjusted and processed according to PyTorch. In 3D GS, parameters such as μ and Σ of the Gaussian body need to be stochastic gradient descending with the help of PyTorch's auto-differentiation framework, in which μ , the color c and the opacity α are more accessible to obtain. However, since PyTorch's auto-differentiation framework needs to show the Σ gradient given when optimizing Σ to ensure the algorithm works appropriately, 3D GS has been re-parameterized Σ . The expression is as follows. The specific expression is shown below:

$$\sum = RSS^T R^T \quad (3)$$

3.2.3 Splatting of Gaussian

Traditional rasterization is a crucial step in 3D graphics rendering. Its main task is to convert geometric shapes (commonly triangles) in 3D space into pixels on a 2D screen and colorize them to produce the final image. In modern GPU pipelines, the hardware typically automatically performs the rasterization process, so 3D GS must design and optimize the CUDA kernel in-house to implement the rasterization procedure. The 2D graphic obtained by 3D GS after projecting a Gaussian body onto the projection plane is called Splatting. As Fig. 6 shows, the assumption is that the origin of the camera coordinate system is located at the center of the projection, and the coordinates of the camera in space are $\mathbf{t} = (t_0, t_1, t_2)^T$, where the dimension t_2 is located is the z-axis and refers to the lens orientation; after the transformation, the camera coordinates are $\mathbf{x} = (x_0, x_1, x_2)^T$.

Equation (3) shows the transformation process of the camera space and the ray space. During the transformation process, 3D GS must determine whether the rays are in contact with the Gaussian body. 3D GS does not use an iterative approach for each ray in different directions, which simplifies the entire computational process.

$$\mathbf{x} = (x_0, x_1, x_2)^T = \phi(\mathbf{t}) = \begin{pmatrix} \frac{t_0}{t_2} \\ \frac{t_1}{t_2} \\ \|(t_0, t_1, t_2)^T\| \end{pmatrix} \quad (4)$$

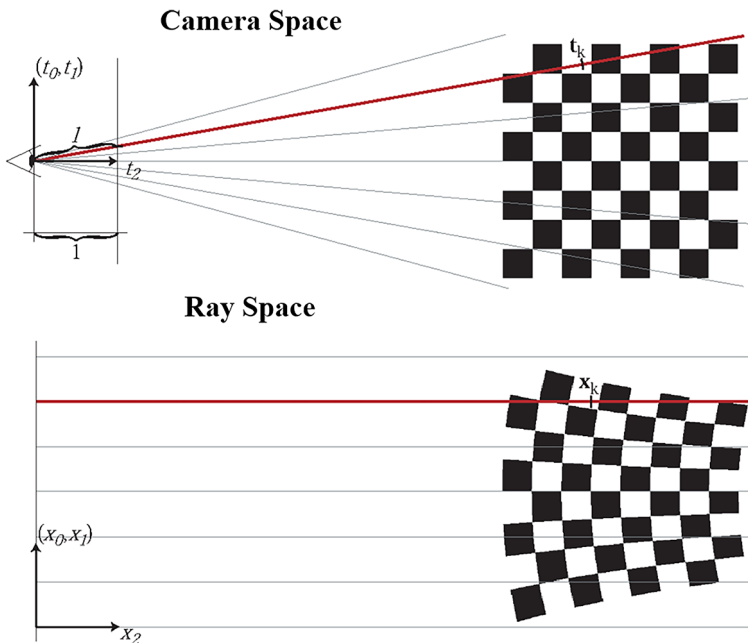


Fig. 6 Schematic diagram for converting volumes from camera space to ray space (Zwicker et al. 2001)

The mean value of the Gaussian function is known for the 3D GS at each projection so that the local affine mapping approximation ϕ_k can be defined as the first two terms of the Taylor expansion of the Gaussian center point $\mathbf{t}_k$ in camera space. The Jacobi matrices appearing in the process will naturally provide a linear transformation from the $\mathbf{t}_k$ neighborhood to the $\mathbf{x}_k$ neighborhood, the expression of which is shown in Eq. (4). The partial derivation of ϕ at point $\mathbf{t}_k$ gives the Jacobian factor $\mathbf{J}_k$. $\mathbf{J}_k$ is also fixed uniquely when given a Gaussian body in camera space with coordinates $\mathbf{t}_k$. Equation (5) shows the expression.

$$\phi_k(\mathbf{t}) = \mathbf{x}_k + \mathbf{J}_k \cdot (\mathbf{t} - \mathbf{t}_k) \quad (5)$$

$$\mathbf{J}_k = \frac{\partial \phi}{\partial \mathbf{t}}(\mathbf{t}_k) = \begin{pmatrix} \frac{1}{t_{k,2}} & 0 & -\frac{t_{k,0}}{t_{k,2}^2} \\ 0 & \frac{1}{t_{k,2}} & -\frac{t_{k,1}}{t_{k,2}^2} \\ \frac{t_{k,0}}{l'} & \frac{t_{k,1}}{l'} & \frac{t_{k,2}}{l'} \end{pmatrix}, \quad l' = \|(t_{k,0}, t_{k,1}, t_{k,2})^T\| \quad (6)$$

3D GS utilizes second-order approximation to solve the nonlinearity problem of projection transformation from 3D space to 2D screen. In the whole processing flow, by integrating the normalized 3D Gaussian $\mathcal{G}_V^3$ along a particular coordinate axis, the normalized 2D Gaussian $\mathcal{G}_V^2$ can be obtained with the expression of Eq. (6). Meanwhile, the covariance matrix of the 2D Gaussian can be obtained directly by removing the third row and column of the 3D Gaussian, which reduces the computational cost. Compared with the traditional volume and integral-based rendering methods (e.g., ray tracing, path tracing, etc.), 3D GS cleverly utilizes the equivalence of 3D Gaussian and 2D Gaussian axial integral, which significantly reduces the spatial dimensions that need to be sampled in terms of the mathematical mechanism, and thus reduces the computational complexity.

$$\int_{IR} \mathcal{G}_V^3(\mathbf{x} - \mu) d_{x_2} = \mathcal{G}_V^2(\hat{\mathbf{x}} - \hat{\mu}) \quad (7)$$

4 3D GS for automatic driving field

The Society of Automotive Engineers (SAE International) in 2014 provided a six-point hierarchy of autonomous driving technologies (Teoh 2020), as shown in Table 2. In Table 2, to realize fully autonomous driving, the onboard system must quickly recover the 3D structure of the surrounding environment (including the shape of the road, lane line position, obstacle information, and dynamics of other vehicles and pedestrians) from the vehicle sensor data in real time. In this manner, the self-driving vehicle can accurately understand the environment and quickly respond to the events. In addition, the continuous release of favorable policies at home and abroad has strongly promoted the development of vehicle intelligence and facilitated the stage-by-stage substantial development of autonomous driving technology. Different states in the U.S. have different regulatory standards for autonomous driving. As of March 2024, some U.S. states (e.g., California, Arizona, and Nevada) have established a relatively comprehensive and complete regulatory framework to support testing autonomous driving vehicles and commercial deployment under specific conditions. Europe is also

Table 2 The society of automotive engineers' grading of automated driving technologies (Teoh 2020)

SAE classification	SAE definitions	Driving operation	Perimeter monitoring	Secure exit	Request response
Level 0 (no automation)	Driver's full authority on driving tasks	Pilot	Pilot	None	None
Level 1 (assisted driving)	The assistance system performs a steering or acceleration/deceleration task in a specific driving mode based on information about the driving environment and relies on the driver to perform all remaining aspects of the dynamic driving task	Pilot+System	Pilot	Pilot	Response
Level 2 (partially automation)	One or more assistive systems that use information about the environment to perform steering, acceleration/deceleration in specific driving modes and rely on the driver to perform all other aspects of dynamic driving tasks	System	Pilot	Pilot	Responsive
Level 3 (conditional automation)	The automated driving system accomplishes all aspects of the dynamic driving task and expects the driver to respond appropriately when requested to intervene	System	Unmanned System	Receivers	Responsive
Level 4 (highly automation)	Autonomous driving systems accomplish all aspects of dynamic driving tasks and request intervention periods where the driver is not necessarily expected to respond	System	Unmanned System	System	Unavoidable responsive
Level 5 (fully automation)	Autonomous driving systems perform all aspects of dynamic driving tasks in all road and environmental conditions that can be managed by a human driver	System	Unmanned System	System	None

optimistic about self-driving technology. Germany, France, and Sweden have a robust automotive industry base and actively promote the renewal of intelligent models. The United Kingdom and Finland represent countries that actively promote the development of regulations and standards for self-driving. On November 17, 2023, The Ministry of Industry and Information Technology, Ministry of Public Security, Ministry of Housing and Urban–Rural Development, and Ministry of Transportation of China jointly issued a circular on the pilot work on the admission and on-road passage of intelligent networked vehicles. With the booming autonomous driving technology, the scene reconstruction technology represented by 3D GS plays a crucial role in ensuring the safe and reliable driving of autonomous vehicles. By accurately perceiving and rapidly modeling the surrounding environment, 3D GS can provide rich environmental information for the autonomous driving system. It can help the vehicle better understand the surrounding environment and make safer and more effective driving decisions. Various studies (Chen and Wang 2024; Fei et al. 2024; Wu et al.

2024a; Tosi et al. 2024) have reviewed and summarized the current application of 3D GS. However, this literature covers a wide range and does not target autonomous driving. This section introduces the application aspects of 3D GS-based 3D scene reconstruction technology in autonomous driving, which contains three main aspects: new perspective synthesis, scene understanding, and SLAM.

4.1 New perspective synthesis

Autonomous driving systems collect data from multiple sensors, which are often from different viewpoints and angles. A technique to fuse the data from multiple sources to create a unified representation of the 3D scene and provide new viewpoint information is required to help the autonomous driving system more accurately perceive and understand the surrounding environment. The new perspective synthesis technique converts a 3D scene to a new perspective using 3D reconstruction algorithms (e.g., surface reconstruction and body mapping) to create a 3D model of the surrounding environment and convert it to a new perspective. In autonomous driving, the commonly used methods to synthesize new perspectives are Bird's-Eye View (BEV)-based methods (Li et al. 2023a; Cai et al. 2023), Transformer-based methods (Wang et al. 2021b; Suhail et al. 2022), and NeRF-based methods (Mueller et al. 2023; Yang et al. 2023c). However, in the practical application process, these methods have insufficient detail restoration, poor real-time performance, and high computational resource requirements. Vehicles must accurately identify potentially dangerous targets during driving, respond to emergencies in real time, and minimize the cost price of on-board equipment. Thus, 3D GS has injected new vitality into the research of new perspective synthesis with its efficient point cloud processing speed, robust surface reconstruction process, real-time data processing performance, and flexible viewpoint conversion process.

When synthesizing new perspectives, the main challenge is to accurately model and render complex optical properties of reflective surfaces by considering how the reflective surfaces change their reflective content with the changes in viewing angle. The 3D GS model initializes the Gaussian body from a sparse point cloud generated by the SfM and controls the density of the Gaussian body by densification and pruning of the points, thus better capturing the details of the reflective surfaces of objects in the scene. However, reflective surfaces produce high dynamic range lighting effects and reflect the details of the surroundings, so the rendering algorithms must be able to handle complex interactions of light with the surface, including reflection, refraction, and scattering. Discrete raw 3D Gaussian models cannot accurately capture these details and cause distorted reflection effects. To address the rendering problem of scenes with reflective surfaces, especially with discrete 3D Gaussian models, Jiang et al. (2023) proposed a novel model called GaussianShader. As is shown in Fig. 7, GaussianShader proposes a new standard estimation framework based on the shortest axis direction of the 3D Gaussian model and designs a detailed loss function to maintain consistent geometries of the normal and Gaussian sphere to achieve good progress and balance between efficiency and visual quality. Some researchers have noted that more details of the scene geometry, surface materials, and ambient lighting in multi-view images can be captured from unknown lighting conditions through innovative rendering approaches. Liang et al. (2023) introduced 3D GS to inverse rendering and proposed a model GS-IR. GS-IR combines a depth-derivative-based regularization method and a baking-based occlusion method to enable a faster and more compact geometric reconstruction and a realistic and

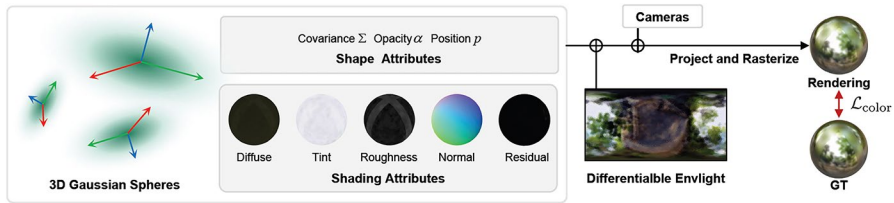


Fig. 7 The overview of GaussianShader (Jiang et al. 2023)

novel synthesis of new views. GS-IR greatly solves the problems of low expressive capacity and high computational demands faced by MLPs in NeRF, improving the effectiveness and efficiency of inverse rendering, especially when it cannot be rendered at interactive rates. Yang et al. (2023d) proposed the Spec-Gaussian model, which started by improving the spherical harmonic function (SH) computation. Spec-Gaussian uses an anisotropic spherical Gaussian (ASG) appearance field for scene reconstruction, which significantly improves the modeling effectiveness and capability of 3D GS for scenes with specular reflection and anisotropic components.

For autonomous driving, improving the performance of monocular and small lens reconstruction techniques can significantly reduce the overall cost of an autonomous driving system, improve the efficiency and reliability of the overall perception system, and is an important research direction for new perspective synthesis. However, the main constraints of monocular and small-lens reconstruction techniques are the presence of mutual occlusion or lack of texture of scene objects, variations in lighting conditions and dynamic scenes, and uncertainty of the absolute scale of the scene (Duan et al. 2021). Charatan et al. (2023) proposed a forward model called pixelSplat to reconstruct the 3D radial field from image pairs using 3D Gaussian primitives, which addressed the problem of wide-angle new viewpoint synthesis. In the training process, pixelSplat implements the back propagable Gaussian distribution sampling through predictive probability distributions and differentiable reparameterization techniques; thus, pixelSplat solves the local minima problem that is inherent in sparse and locally supported representations. PixelSplat, which brings the benefits of a primitive-based 3D representation—fast and memory-efficient rendering as well as interpretable 3D structure—to generalizable view synthesis. Chen et al. (2024a) proposed that unlike the existing data-driven design methods, a cost-volume-driven approach could utilize the multiple-view correspondence information to improve the geometry learning. Chen et al. (2024a) proposed a model called MVSplat. Compared to pixelSplat, MVSplat sacrificed some of the reflective surface computational performance, but could predict a reliable probabilistic depth distribution solely from image features. However, MVSplat uses 10 times fewer parameters, improves the inference speed by more than 2 times, and provides higher appearance, geometric quality, and better generalization across datasets.

Some researchers have combined the explicit structural features of 3D GS and the excellent properties of other methods to continuously break the records of various indices in new view synthesis. Zou et al. (2023) proposed a new hybrid Triplane-Gaussian 3D representation model, which organically combined explicit and implicit features. Triplane-Gaussian could achieve the generalizable 3D reconstruction using single-view images by introducing local image spatial features in the Transformer backbone network to introduce the local image spatial features for conditional control. Yang et al. (2023b) proposed Deformable 3D

Gaussians as a monocular dynamic scene modeling pipeline model based on rasterization. By combining 3D GS, deformation field techniques, and an annealing smoothing training mechanism, Deformable 3D Gaussians performs well on new view synthesis, temporal interpolation, and real-time rendering of the SOTA results on multiple datasets.

4.2 Scenario understanding

The new perspective synthesis task aims to fuse the existing data to create a unified representation of a 3D scene, so that vehicles can accurately understand the current environment. Meanwhile, scene understanding is to analyze, interpret, and reason about the environment information to identify and understand the basic properties, associative relationships, and overall layout of the objects in the scene (Abdul et al. 2022). This process includes tasks such as object detection, semantic segmentation, instance segmentation, scene classification, and action recognition. Because 3D scene understanding is closer to the actual physical world and provides richer semantic information, its research results have a wide range of application prospects in human-computer interaction. Thus, 3D scene understanding has become one of the important research directions. As shown in Fig. 8, 3D scene understanding can be divided into two categories: "3D+2D" scene understanding and "3D+language" scene understanding (Trabelsi et al. 2022); the major difference is in the methods of information fusion and focus.

"3D+2D" scene understanding combines 3D spatial information with 2D image information to comprehensively interpret and explain the scene. This approach relies on image

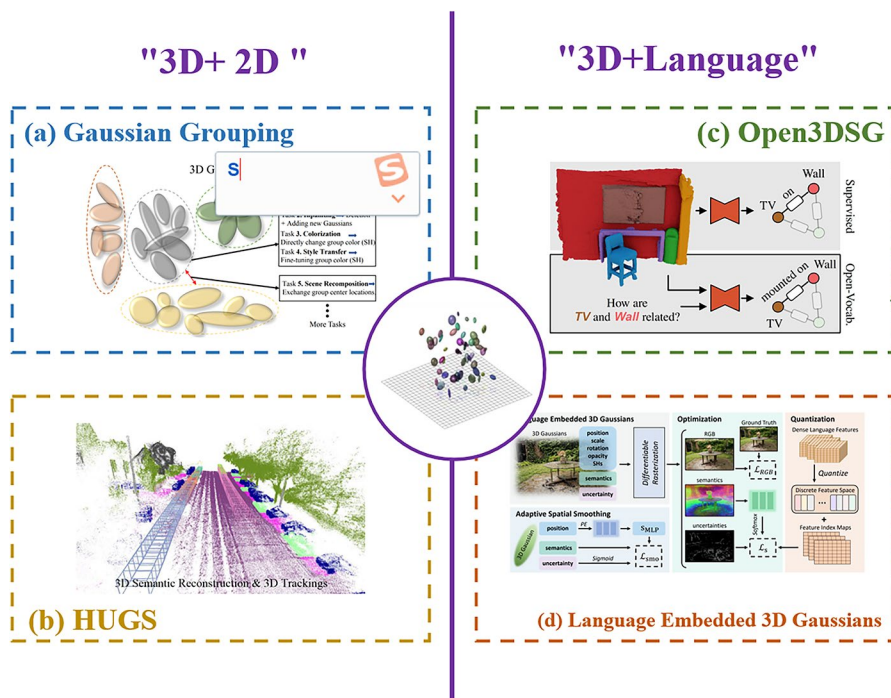


Fig. 8 Schematic diagram of the 3D scene understanding categorization situation (Ye et al. 2023; Zhou et al. 2024b; Sebastian et al. 2024; Shi et al. 2023)

processing and 3D reconstruction techniques to extract the feature information in a 2D image. Meanwhile, it fuses the information with the geometric structure in 3D space to understand the scene in depth. The current mainstream solutions are based on the geometric alignment (Li et al. 2023b) or semantic alignment (Zhang et al. 2023d) to deeply fuse information from LiDAR and the camera. However, the alignment process is limited by various aspects such as sensor calibration errors, dynamic scene changes, and operational arithmetic requirements. Due to the excellent performance of 3D GS, some researchers noted that the fine-grained segmentation capability of the 2D segmentation base model can be refined into 3D GS to avoid the enormous arithmetic and time cost of multiple forward propagations during the inference process of the traditional method. Cen et al. (2023a) organically combined the 2D segmentation base and 3D GS models and proposed a model for use in the millisecond 3D Gaussian with the fine-grained 3D interactive segmentation method SAGA. The core strategy of SAGA is to extract 2D features and masks using the SAM model, train the 3D features to be attached to the Gaussian body, and design the SAM bootstrap loss and corresponding loss function to guide the training of the Gaussian body. Compared with SA3D Cen et al. (2023b), SAGA can get 3D assets with higher quality in much less time. As shown in Fig. 8a, Ye et al. (2023) extended 3D GS to reconstruct and segment open-world scenes and proposed the Gaussian Grouping model. Gaussian Grouping uses the SAM model to supervise the identity encoding of each Gaussian body in differentiable rendering. The encoding can be effectively grouped into a specific scene based on object instances or scene members, which demonstrates the flexibility and efficiency in scene understanding tasks. Lin et al. (2024) proposed VastGaussian as a 3D GS-based method for high-quality reconstruction and real-time rendering of large scenes through group partitioning. VastGaussian uses an incremental data partitioning strategy to achieve parallel optimization and seamless system merging. In addition, it uses a detachable and decoupled appearance modeling module to inhibit the dynamic floating of the model. As shown in Fig. 8b, Zhou et al. (2024b) proposed a novel 3D GS-based model called HUGS based on only RGB images for the overall understanding of the urban scene. HUGS adopts the unicycle model as the basic model to construct a dynamic scene, generates an exposure affine matrix for each camera by mapping the extrinsic parameters of the camera to the affine matrix through a small MLP, and generates an exposure affine matrix for each camera through a small MLP. Each camera generates an exposure affine matrix by mapping the extrinsic parameters of the camera to an exposure affine matrix, which only relies on a pre-trained recognition model (containing noisy 2D semantic and instance predictions, noisy 2D optical flow, and noisy 3D tracking results) to achieve the pure RGB holistic scene understanding.

"3D+Language" scene understanding combines 3D spatial information with natural language information to describe and interpret the scene. This understanding approach can understand the scene in depth by transforming the information in 3D space into natural language descriptions or guiding the understanding of 3D space through natural language descriptions. Current mainstream solutions are based on methods such as Transformer (Zhu et al. 2023b) or graph neural networks (Jin et al. 2021) to realize the multimodal interaction. However, the main drawbacks in training are the high complexity of the model, high demand for computational resources, and high cost of data collection and annotation. As shown in Fig. 8c, Sebastian et al. (2024) proposed Open3DSG to predict 3D scene graphs without labeling the data. Open3DSG, the first interactive graph representation of scene creation based on a 3D point cloud, uses the 2D visual-linguistic model knowledge to pre-

dict open-vocabulary 3D scene graphs via 3D graph neural networks. Although the efficiency bottleneck of existing methods depends on resource-intensive neural networks, the key problem of 3D GS in "3D+Language" scene understanding is how to embed linguistic features without degrading the performance. As shown in Fig. 8d, Shi et al. (2023) proposed a novel scene representation called Language Embedded 3D Gaussians by adopting a specialized quantization scheme and a novel embedding process to solve the language-embedded 3D GS problem. Language Embedded 3D Gaussians successfully embeds quantized and compact semantic features into Gaussian with minimal memory and storage space. Thus, Language Embedded 3D Gaussians achieved excellent visual quality and linguistic query accuracy in the current field of "3D+Language" scene understanding. Compared that 3DOVS (Liu et al. 2023a) and LERF (Justin et al. 2023) shows poor performance, Language Embedded 3D Gaussians supports high frequency embedded semantic features, allowing for enhanced spatial semantic accuracy simply by incorporating more 3D Gaussians into scenes. The GaussCtrl model proposed by Wu et al. (2024b) combines 3D GS with a 2D diffusion model to achieve language-driven 3D scene reconstruction through innovative multi-view consistency editing.

4.3 Simultaneous localization and mapping

SLAM (Hugh et al. 1996) refers to the simultaneous localization and map building, which aims to achieve autonomous localization and map building for intelligent systems such as robots or unmanned vehicles by fusing the sensor data and motion models. VSLAM (Visual SLAM) (Yan et al. 2022) relies on visual information for attitude estimation and map generation. VSLAM has outstanding advantages such as low cost, rich information, and easy integration and is one of the key research directions in SLAM research. The NeRF-based SLAM method uses neural networks to represent the scene radiation field, directly learns the continuous density function in 3D space from the original pixel values, and generates a new perspective through volume rendering techniques (Daniil et al. 2023). Compared with the traditional single-point substitution methods, the NeRF-based SLAM has no manual feature extraction process. It can use a gradient descent to optimize the representation of the scene. However, it has a slower rendering speed, a low localization accuracy, and higher overall noise in practical applications. The 3D GS-based SLAM method continues the advantages of the NeRF-based SLAM method. Meanwhile, it is more efficient in terms of real-time processing, localization accuracy, rendering speed, and result quality to comprehensively improve the performance (Seongbo et al. 2024).

Yan et al. (2023) proposed GS-SLAM, which was the first model to use 3D GS representation in the SLAM domain. GS-SLAM uses a real-time microscopic splatting rendering pipeline to provide significant speedup ratios for map optimization and RGB-D redrawing. GS-SLAM uses an adaptive scaling strategy to improve the build quality and obtains a surprisingly fast rendering speedup in the Improvements in rendering speed. However, regarding the overall framework operation efficiency and localization accuracy, GS-SLAM has not significantly improved. Keetha et al. (2023) proposed SplaTAM, which was a dense RGB-D SLAM solution based on 3D GS. SplaTAM streamlines the architecture of the underlying 3D GS using only view-independent colors and making the Gaussian distribution isotropic. Simultaneously, it uses microscopic rendering to directly compute the gradients of Gaussian bodies and camera parameters. The final model achieves fast rendering at

high accuracy. Hidenobu et al. (2023) proposed the first monocular SLAM model, Gaussian Splatting SLAM, entirely based on 3D GS, which was the first model to achieve real-time incremental 3D GS-based reconstruction at a speed of 3 FPS and uses Gaussians as the only 3D representation. Similarly, other works such as Gaussian-SLAM (Vladimir et al. 2023), Photo-SLAM (Huang et al. 2023), and NEDS-SLAM (Ji et al. 2024) have advanced the development of 3D GS-based SLAM. Table 3 shows the rendering results of some SLAM models on the Replica dataset. The 3D GS-based model significantly improves all indices compared to the implicit neural network model.

Some researchers used multi-source heterogeneous data structures to compensate for the lack of visual information and achieved good results on the broader application level of autonomous driving. Hong et al. (2024) proposed a multi-modal sensor fusion map building system call LIV-GaussMap, which integrated precision LiDAR, IMU, and vision. LIV-GaussMap uses LiDAR and vision data to capture and recover the scene geometry information, IMU data to initialize the surface Gaussian body attitude, and visual luminosity gradient to manage and optimize the Gaussian body. Then, LIV-GaussMap builds an accurate map structure, which can play an essential role in the integrated environment sensing task of autonomous driving.

Research on the dynamic scenes of autonomous driving demonstrates the extreme ability of 3D GS to accurately model multiple independent models and accurately represent the dynamic foreground. Chen et al. (2023b) proposed the representation model PVG, which unifies dynamic and static elements, to introduce periodic vibration-based temporal dynamics into efficient static 3D GS techniques. The model combines a temporal smoothing mechanism and a location-aware adaptive control strategy to enable dynamic large-scale urban scene reconstruction without relying on manually labeled object bounding boxes or expensive optical flow estimation. Zhou et al. (2023) proposed DrivingGaussian, the first composite 3D GS-based representation and modeling framework for large-scale dynamic driving scenes. DrivingGaussian divides the modeling process of a dynamic driving scene into two parts: the incremental static 3D Gaussian module and the composite dynamic Gaussian map module. The complex driving scene is hierarchically modeled based on the input multi-source heterogeneous data order to form a high-efficiency and high-quality framework for the 3D reconstruction and simulation of dynamic surround-view driving scenes. Yan et al. (2024) proposed that the Street Gaussians model also adopted a layered modeling approach, but they used a 4D ball-harmonic model in the dynamic foreground modeling to improve the lack of accuracy of the original 3D GS in simulating moving vehicles. The model generates

Table 3 Rendering effectiveness of different SLAM methods on the Replica (Straub et al. 2019) dataset (Chen and Wang 2024; Wu et al. 2024a)

Method type	Model	Norm		
		PSNR↑	SSIM↑	LPIPS↓
Neural network model	NICE-SLAM (Zhu et al. 2022)	24.42	0.81	0.23
	Vox-Fusion (Yang et al. 2022)	24.41	0.80	0.24
	Co-SLAM (Wang et al. 2023)	30.24	0.94	0.25
3D GS model	GS-SLAM (Mueller et al. 2023; Yan et al. 2023)	31.56	0.97	0.094
	SplaTAM (Keetha et al. 2023; Yang et al. 2023c)	34.11	0.97	0.10
	Gaussian Splatting SLAM (Hidenobu et al. 2023; Suhail et al. 2022)	37.50	0.96	0.07
	Gaussian-SLAM (Chen et al. 2024b; Yan et al. 2023)	38.90	0.99	0.07

high-quality renderings at 133 FPS (1066×1600 resolution) in 30 min with only estimated trajectories.

5 Future research directions

Although 3D GS-based scene reconstruction techniques have achieved significant success in the autonomous driving-oriented field, including their high accuracy, fast rendering, and high-quality features, there is still much room for improvement. For example, 3D GS requires colossal storage space when expressing complex details. The projection method of 3D GS will exacerbate problems such as aliasing and artifacts, and inadequate initialization will cause the overfitting and over-smoothing of subsequent models. Figure 9 shows a brief framework schematic in Chapter 5 to focus on the research frontiers of 3D GS, which is an innovative technology in autonomous driving, including structural optimization, 4D scene reconstruction, and vehicle navigation aspects.

5.1 Structural optimization

Structural optimization is a crucial method to improve the performance and efficiency of any algorithm. Although 3D GS techniques can effectively separate ground points from non-ground points, their accuracy and robustness are affected by occlusions, lighting conditions, and fast-moving objects. In addition, to satisfy the real-time requirements, 3D GS technology must reduce the dependence on computational resources and increase the processing speed. The structural optimization of 3D GS can be enhanced by optimizing the Gaussian body management, streamlining the central body architecture, and eliminating artifacts to improve its performance. The structural optimization can make 3D GS more accurate, efficient, and reliable, so that 3D GS can better adapt to various complex and realistic driving scenarios.

In 3D GS, Gaussian bodies are data structures to represent the color and depth of pixels in a scene. They simulate objects and surfaces in a scene by distributing Gaussian functions in 3D space. Controlling and optimizing the distribution and management of Gaussians can effectively improve the rendering efficiency and quality. As shown in Fig. 10, in the

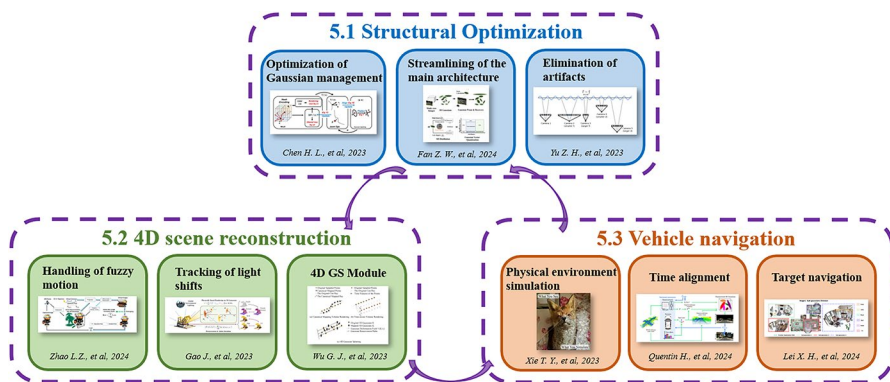


Fig. 9 Schematic diagram of the framework for Chapter V

NeuSG model proposed by Chen et al. (2023c), the center of the Gaussian body is guided to approach the surface by adding scale regularization. Meanwhile, NeuSG uses a neural implicit model to refine the point cloud of the Gaussian body, which generates a completely reconstructed surface with complex details. In the SuGaR model proposed by Antoine and Vincent (2023), the Gaussian body is also encouraged to align with the scene surface by introducing a regularization term. The mesh is extracted from the Gaussian function using Poisson reconstruction. Finally, editable rendering is performed by binding the Gaussian body to the surface of the mesh, which achieves faster results with better rendering quality. In the Octree-GS model, Ren et al. (2024) noted that an excess of Gaussian bodies in the view cone would limit the real-time rendering of 3D GS for large scenes with complex details. Therefore, they introduced the LOD structure to manage and organize the Gaussian bodies and better obtained the scene details using a new adaptive tuning scheme and progressive training strategy. Yu et al. (2024) combined the advantages of the flexible and efficient 3D GS and the neural symbolic distance field SDF using the three inter-guided approaches (geometric pattern density control, mutual geometric supervision, and depth-guided ray sampling mechanisms) to enhance the efficiency of the Gaussian body management and facilitate the joint performance.

Streamlining of the leading architecture refers to optimizing and simplifying the system architecture of the three main modules of 3D GS: 3D Gaussian representation module, attribute optimization module, and real-time rendering module. This streamlining process aims to improve the system efficiency and performance, so that 3D GS can play a more important role in dynamic scene reconstruction, mobile carrier devices, and interactive applications. Liu et al. (2023b) addressed the problem of real-time rendering for 3D reconstruction on mobile devices by adopting a dual-sputtering architecture to reduce the computational cost and reduce the processing of low-contributing points to minimize redundant data generation. The model introduces a thread saturation overflow operation to speed up the overall convergence rate and uses voxelized point clouds to reduce the rendering and training overhead. Ultimately, for large-scale scene datasets, the improved method can reduce the model space occupation by 37.08% on average while ensuring the rendering quality. Fan et al. (2023) proposed the LightGaussian model using the network pruning idea, which VecTree quantized through the identification of determination, pruning restoration, distillation degradation, and process. This method maintains the quality of the reconstruction while dramati-

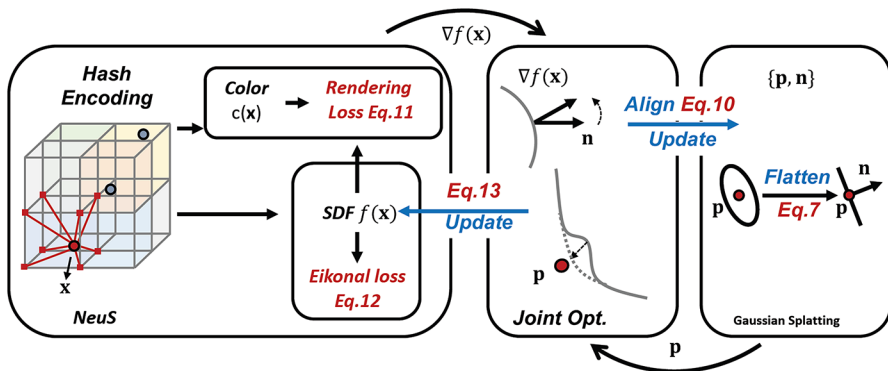


Fig. 10 The NeuSG framework (Chen et al. 2023c)

cally reducing the storage overhead. Ultimately, LightGaussian could achieve an average $15\times$ compression rate, and the FPS can improve from 139 to 215. Stanislaw et al. (2023) proposed Splatter Image, a learning-based method for ultra-fast monocular 3D object reconstruction. Splatter Image uses a 2D image-to-image network to map the input image to a 3D Gaussian body at each pixel. Splatter Image only required the feedforward evaluation of the neural network to reconstruct it, and its maximal operational speed is 38 FPS.

3D GS may have multi-view inconsistencies between camera view and training view, which are known as artifacts, during the synthesis of the new view. The presence of artifacts will cause difficulties in reconstructing the 3D GS model in some regions (e.g., underdense and overclipped) (Tang et al. 2023). Yu et al. (2023) attributed this phenomenon to using a 2D dilation filter and the lack of 3D frequency constraints. Their proposed Mip-Splatting model introduces a 3D smoothing filter and replaces the 2D2D expansion filter with a 2D Mip filter that mimics a 2D box filter. Wang et al. (2024) introduced the view-consistent editing VcEdit framework, which attempted to alleviate the pattern collapse and visual artifacts in image editing to guide the multi-view consistency. The final model effectively solves the multi-view inconsistency problem and realizes high-quality 3D GS editing in various scenes.

For the 3D GS technology for autonomous driving, the improvement of its editing and maneuverability will also benefit the system by more accurately tracking the environmental targets and more flexibly handling potential risks. However, more methods to accurately recognize Gaussian targets are required, and the mobility of 3D GS causes its poor convergence (Lei et al. 2024a; Chen et al. 2023d). To address these problems, Chen et al. (2024b) proposed a novel, fast, and highly controllable 3D Gaussian editing algorithm called GaussianEditor. GaussianEditor introduces Gaussian semantic tracking into 3D GS, adopts a hierarchical Gaussian splicing method to guide the highly stochastic generation, and designs a 3D repair method based on 3D GS.

5.2 4D scene reconstruction

Based on the success and potential of 3D GS for static scene reconstruction and new perspective synthesis, researchers could extend the application of 3D GS to dynamic scenes to achieve high-level autonomous driving in real time. 4D scene reconstruction techniques require high-quality reconstruction of 3D spatial structures by in-vehicle systems and must consider variations in the temporal dimension to capture the complex dynamics of rigidly moving objects. Existing approaches commonly rely on learning a canonical space and encoding the dynamic changes as one of the implicit or explicit deformation fields, i.e., each point in a dynamic scene is mapped to a static reference frame to capture its motion over time. However, the rapid motion of the target object will drastically change the deformation field, so the reconstructed model cannot accurately reflect the real dynamic scene. Zhao et al. (2024) introduced 3D GS into the processing of motion-blurred images and proposed a model called BAD-Gaussians. BAD-Gaussians introduced the first photometric beam adjustment formula for motion-blurred images, which could render motion-blurred images in real time within the 3D GS stitching framework. Gao et al. (2024) proposed the GaussianFlow model by linking the 3D Gaussian dynamics between consecutive frames to pixel velocities and obtained the Gaussian flow by stitching the Gaussian dynamics in the

image space. GaussianFlow solves the problems of 4D scene dynamics monitoring, 4D new view synthesis, and Gaussian body color drift well.

4D scene reconstruction techniques require models with strong tracking ability for light transformations during motion. In contrast, the basic 3D GS models focus on the geometric information of the ground and lack the material and lighting attribute information of the scene, so they cannot be used to reconstruct a scene that can be relighted. Gao et al. (2023) developed an integrated rendering pipeline based on the 3D Gaussian representation Relightable 3D Gaussian, which can support relighting, editing, and ray tracing to reconstruct 3D point clouds and achieve real-time rendering levels.

Meanwhile, some researchers have proceeded to directly model 4D GS. Although the specific research methods vary, introducing the same scale to harmonize the dynamics between 3D GS and timestamps is the core idea. As shown in Fig. 11, Wu et al. (2023) used a compact representation and an efficient deformation field to model the motion. Then, they splashed the Gaussian body onto the timestamped image, which ultimately built the same 4D GS model. Duan et al. (2024) proposed a model 4D GS based on anisotropic 4D XYZT Gaussians to reconstruct dynamic scenes, and the main idea is to directly apply the time slicing method to dynamic 3D GS. Yang et al. (2023e) proposed a 4D GS model where the spacetime was viewed as a whole and modeled by a rotating 4D Gaussian body that is anisotropic: arbitrarily available in time and space. This 4D Gaussian body is represented by 4D spherical harmonic coefficients to approximate the underlying spacetime 4D volume of the dynamic scene, which achieves efficient real-time rendering. In contrast, Jeongmin et al. (2024) noted the erroneous design of deformation fields in the existing dynamic studies. They proposed defining the deformation as a function of per-Gaussian and temporal embedding, which outperformed the 4D GS model on multiple datasets.

Besides all of the above, how to significantly elevated scene reconstruction efficiency and novel view synthesis (NVS) accuracy without camera parameters provided by COLMAP, particularly for dynamic scenes, is one of the important issues 4D Scene Reconstruction needs to consider.

5.3 Vehicle navigation

Navigation technology can provide the necessary location information and driving routes for vehicles, which are among the vital essential services to realize high-level autonomous driving. Based on the excellent performance demonstrated by 3D GS in scene reconstruction of automatic driving, extending 3D GS to the research on automatic driving navigation

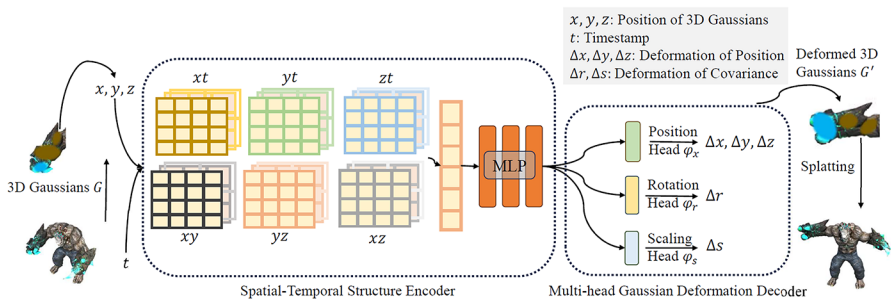


Fig. 11 The overall pipeline of 4D Gaussian Splatting (Wu et al. 2023)

can effectively break the bottleneck of the quality and efficiency of the existing navigation methods and significantly improve the accuracy and reliability of automatic driving navigation.

When the in-vehicle system can perform a realistic physical simulation of the world environment, the vehicle navigation will be able to simulate and predict the changes in the surrounding environment and more intelligently perceive, plan, and make decisions. The physical simulation of the world environment can significantly enhance the safety and reliability of vehicle navigation and enable vehicles to flexibly respond to various emergencies in complex and changing traffic environments, such as changes in road conditions, shifts in the surrounding environment, and the moving intentions of dynamic targets. Xie et al. (2023) seamlessly integrated physics-based Newtonian dynamics with 3D GS and proposed the novel model PhysGaussian for effective physics simulation. PhysGaussian employs a particular material-pointing method to incorporate the mechanical properties into the Gaussian body, eliminate the necessity of geometric embedding, and exhibits excellent physics simulation performance.

In vehicle navigation, accurately obtaining the spatio-temporal alignment from multimodal fusion data is one of the important research directions. Traditional methods require artificial markers and have poor generalization ability, so a reliable and trustworthy targetless calibration method is critical in vehicle calibration. Quentin et al. (2024) proposed a 3D GS-based calibration method, 3DGS-Calib, which has faster calibration speed and higher calibration accuracy than the existing implicit neural methods and can efficiently complete the in-vehicle multimodal spatio-temporal calibration task.

Currently, the more mature visual navigation technology research focuses on map representation methods represented by Bird's Eye View (BEV) and Occupancy Network, which can provide more detailed environment modeling and more accurate environment understanding. However, BEV 2D maps are insufficient to characterize the complex textures and detailed features of a scene, and the Occupancy Network model places severe requirements on the arithmetic platform of the vehicle system. Thus, exploring an applicable, simple, and efficient algorithmic framework for target navigation is one of the common goals of researchers in automated navigation. The proposed framework enables the agent to not only memorize the geometry and semantic information of the scene, but also retain the textural features of objects. As shown in Fig. 12, Lei et al. (2024b) proposed GaussNav, a 3D GS-based instance image target navigation model. GaussNav transforms the original instance image target navigation task into a relatively easy-to-handle point target navigation task. GaussNav uses the three main phases of sub-Gaussian segmentation, semantic Gaussian construction, and Gaussian navigation to efficiently localize the target object and significantly improve the performance.

6 Conclusions

This paper provided a comprehensive review of the research on 3D GS-based 3D scene reconstruction technology for autonomous driving, described the basic situation of 3D GS and its application in autonomous driving, and covered the current latest research results. First, the research background of 3D GS was summarized. Second, the reasons for the high performance of 3D GS were discussed by exploring the mathematical mechanism of the

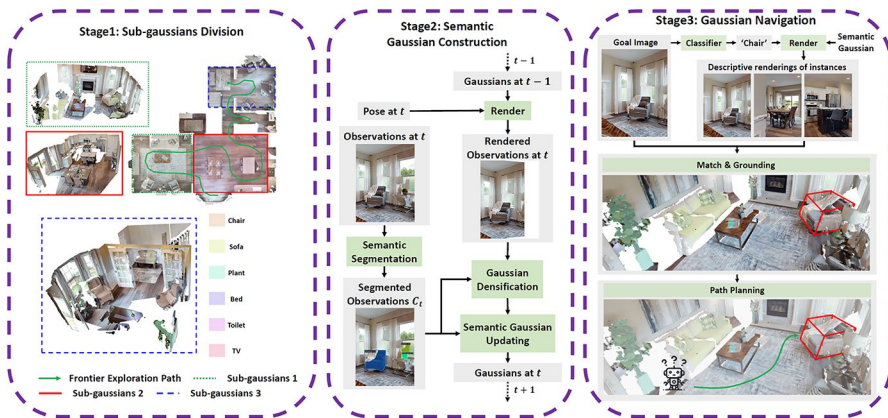


Fig. 12 The three stages of GaussNav framework (Lei et al. 2024b)

core formula of 3D GS as the main body. Third, the three main application directions of 3D GS in autonomous driving were presented: new perspective synthesis, scene understanding, and SLAM, with the emphasis on the revolutionary research trend that 3D GS introduced to autonomous driving. Finally, an in-depth discussion of the issues and research frontiers of 3D GS was given. This targeted overview of the fundamentals, development, and application of 3D GS as an emerging 3D scene reconstruction technology in autonomous driving could provide a reliable channel for researchers to quickly understand and familiarize themselves with 3D GS and build a broader vision thereof, thereby developing ideas for related scientific research.

Author contributions H.X.Z, Z. L. Z and J.Y. Z had the lead in writing the manuscript and carried out most of the background research. H.D helped to find more references and edited and polished the draft. Y.D edited the paper, and helped to define the scope and structure of the survey. X.W.X was also engaged in the process and helped to draft the response to reviewers' comments. J.S.Y prepared most of the figures and the Bibliography Section. All authors reviewed the manuscript.

Data availability This manuscript does not report data generation, and all data mentioned in this manuscript are noted citing sources. This manuscript does not have any research data outside the submitted manuscript file.

Declarations

Conflict of interest The authors declare no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- Abdul M, Sumaira K, Samabia T, Amina J (2022) A fast panoptic segmentation network for self-driving scene understanding. *Comput Syst Sci Eng* 43(10):27–43. <https://doi.org/10.32604/csse.2022.022590>
- Albert P, Enric C, Gerard PM, Francesc MN (2021) D-NeRF: Neural radiance fields for dynamic scenes. In: IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 10318–10327
- Anjos DRK, Ribeiro CS, Lopes DS, Pereira JM (2018) Stroke-based splatting: an efficient multi-resolution point cloud visualization technique. *Vis Comput* 34(10):1383–1397. <https://doi.org/10.1007/s00371-017-1420-7>
- Antoine G, Vincent L. (2023) SuGaR: surface-aligned Gaussian splatting for efficient 3D mesh reconstruction and high-quality mesh rendering. arXiv preprint [arXiv:2311.12775](https://arxiv.org/abs/2311.12775)
- Bao WX, Komatsu R, Asama H, Yamashita A (2023) Applying albedo estimation and implicit neural representations to well-posed shape from shading. *IEEE Access* 11:40038–40048. <https://doi.org/10.1109/ACCESS.2023.3269286>
- Barron JT, Mildenhall B, Verbin D, Srinivasan PP, Hedman P (2022) Mip-NeRF 360: Unbounded anti-aliased neural radiance fields. In: IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 5470–5479
- Cai HX, Zhang ZY, Zhou ZY, Li ZY, Ding WB, Zhao JH (2023) BEVFusion4D: learning LiDAR-camera fusion under bird's-eye-view via cross-modality guidance and temporal aggregation. arXiv preprint [arXiv:2303.17099](https://arxiv.org/abs/2303.17099)
- Cen JZ, Fang JM, Yang C, Xie LX, Zhang XP, Shen W, Tian Q (2023) Segment any 3D Gaussians. arXiv preprint [arXiv:2312.00860](https://arxiv.org/abs/2312.00860)
- Cen JZ, Zhou ZW, Fang JM, Yang C, Shen W, Xie LX, Jiang XP, Tian Q (2023) Segment anything in 3d with nerfs. In: Proceedings of the 37th international conference on neural information processing systems (NeurIPS), pp 25971–25990
- Chabra R, Straub J, Sweeney C, Newcombe R, Fuchs H (2019) StereoDRNet: dilated residual StereoNet. In: IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 11786–11795
- Chang JR, Chen YS (2018) Pyramid stereo matching network. In: IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 5410–5418
- Charatan D, Li SZ, Tagliasacchi A, Sitzmann V (2023) pixelSplat: 3D Gaussian splats from image pairs for scalable generalizable 3D reconstruction. arXiv preprint [arXiv:2312.12337](https://arxiv.org/abs/2312.12337)
- Chen XZ, Song ZH, Zhou J, Xie D, Lu JF (2023a) Camera and LiDAR fusion for urban scene reconstruction and novel view synthesis via voxel-based neural radiance fields. *Remote Sens* 15(18):4628. <https://doi.org/10.3390/rs15184628>
- Chen GK, Wang WG (2024) A survey on 3D Gaussian splatting. arXiv preprint [arXiv:2401.03890](https://arxiv.org/abs/2401.03890)
- Chen YR, Gu C, Jiang JZ, Zhu XT, Zhang L (2023). Periodic vibration Gaussian: dynamic urban scene reconstruction and real-time rendering. arXiv preprint [arXiv:2311.18561](https://arxiv.org/abs/2311.18561)
- Chen HL, Li C, Lee GH (2023) NeuSG: neural implicit surface reconstruction with 3D Gaussian splatting guidance. arXiv preprint [arXiv:2312.00846](https://arxiv.org/abs/2312.00846)
- Chen ZL, Wang F, Wang YK, Liu HP (2023) Text-to-3D using Gaussian splatting. arXiv preprint [arXiv:2309.16585](https://arxiv.org/abs/2309.16585)
- Chen YD, Xu HF, Zheng CX, et al. (2024) MVSPat: efficient 3D Gaussian splatting from sparse multi-view images. arXiv preprint [arXiv:2403.14627](https://arxiv.org/abs/2403.14627)
- Chen YW, Chen ZL, Zhang C, et al. (2024) GaussianEditor: swift and controllable 3D editing with Gaussian splatting. In: IEEE conference on computer vision and pattern recognition (CVPR), pp 21476–21485
- Christoph L, Michael Z (2021) Pulsar: efficient sphere-based neural rendering. In: IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 1440–1449
- Dai A, Niessner M, Zollhofer M, Izadi S, Theobalt C (2017) BundleFusion: real-time globally consistent 3D reconstruction using on-the-fly surface reintegration. *ACM Trans Graph* 36(4):1–18. <https://doi.org/10.1145/3054739>
- Daniil L, Connor H, Steven W (2023) Towards open world NeRF-based SLAM. In: Presented at conference on robots and vision (CVR), pp 37–44
- Duan ZM, Chen YW, Yu HJ, Hu BW, Chen C (2021) RGB-Fusion: monocular 3D reconstruction with learned depth prediction. *Displays* 70:102100. <https://doi.org/10.1016/j.displa.2021.102100>
- Duan YX, Wei FY, Dai QY, He YH, Chen WZ, Chen BQ (2024) 4D Gaussian splatting: towards efficient novel view synthesis for dynamic scenes. arXiv preprint [arXiv:2402.03307](https://arxiv.org/abs/2402.03307)
- Fan JC, Chen M, Mo JQ, Wang SG, Liang QH (2022) Variational formulation of a hybrid perspective shape from shading model. *Vis Comput* 38(4):1469–1482. <https://doi.org/10.1007/s00371-021-02081-x>
- Fan ZW, Wang K, Wen KR, Zhu ZH, Xu DJ, Wang ZY (2023). LightGaussian: unbounded 3D Gaussian compression with 15× reduction and 200+FPS. arXiv preprint [arXiv:2311.17245](https://arxiv.org/abs/2311.17245)

- Fei B, Xu JY, Zhang R, Zhou QY, Yang WD, He Y (2024) 3D Gaussian as a new vision era: a survey. arXiv preprint [arXiv:2402.07181](https://arxiv.org/abs/2402.07181)
- Fridovich KS, Yu A, Tancik M, Chen QH, Recht B, Kanazawa A (2022) Plenoxels: radiance fields without neural networks. In: IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 5501–5510
- Fu Y, Liu SF, Kulkarni A, Kautz J, Efros AA, Wang XL (2023). COLMAP-Free 3D Gaussian splatting. arXiv preprint [arXiv:2312.07504](https://arxiv.org/abs/2312.07504)
- Furukawa Y, Hernandez C (2013) Multi-view stereo: a tutorial. *Found Trends Comput* 9(1–2):1–148. <https://doi.org/10.1561/06000000052>
- Gao J, Gu C, Lin YT, Zhu H, Cao X, Zhang L, Yao Y (2023) Relightable 3D Gaussian: real-time point cloud relighting with BRDF decomposition and ray tracing. arXiv preprint [arXiv:2311.16043](https://arxiv.org/abs/2311.16043)
- Gao QK, Xu QG, Cao Z, et al. (2024). GaussianFlow: splatting Gaussian dynamics for 4D content creation. arXiv preprint [arXiv:2403.12365](https://arxiv.org/abs/2403.12365)
- Garcia S, Pages R, Berjon D, Moran F (2015) Textured splat-based point clouds for rendering in handheld devices. In: International ACM symposium on 3D web technologies (Web3D), pp 227–230
- He XL, Chen JY, Peng SD, Huang D, Li YG, Huang XS, Yuan C, Ouyang WL, He T (2024) GVGEN: Text-to-3D generation with volumetric representation. arXiv preprint [arXiv:2403.12957](https://arxiv.org/abs/2403.12957)
- Hiddenobu M, Riku M, Paul HJK, Andrew JD (2023) Gaussian splatting SLAM. arXiv preprint [arXiv:2312.06741](https://arxiv.org/abs/2312.06741).
- Hong S, He JJ, Zheng XH, Wang HS, Fang H, Liu KC, Zheng CR, Shen SJ (2024) LIV-GaussMap: LiDAR-inertial-visual fusion for real-time 3D radiance field map rendering. arXiv preprint [arXiv:2401.14857](https://arxiv.org/abs/2401.14857)
- Huang HJ, Li LW, Cheng H, Yeung SK (2023) Photo-SLAM: real-time simultaneous localization and photo-realistic mapping for monocular, stereo, and RGB-D cameras. arXiv preprint [arXiv:2311.16728](https://arxiv.org/abs/2311.16728).
- Hugh DW, David R, Eduardo N (1996) Localization of automatic guided vehicles. In: The international symposium on robotics research (ISRR), pp 613–625
- Ivo RF, Vidal CA, Cavalcante NJB (2020) Improved silhouette rendering and detection of splat-based models. *Comput Graph-UK* 93:39–50. <https://doi.org/10.1016/j.cag.2020.09.010>
- Jeongmin B, Seoha K, Youngsik Y, Hahyun L, Gun B, Uh YJ (2024) Per-Gaussian embedding-based deformation for deformable 3D Gaussian splatting. arXiv preprint [arXiv:2404.03613](https://arxiv.org/abs/2404.03613)
- Ji YM, Liu Y, Xie GH, Ma BY, Xie ZW (2024) NEDS-SLAM: a novel neural explicit dense semantic SLAM framework using 3D Gaussian splatting. arXiv preprint [arXiv:2403.11679](https://arxiv.org/abs/2403.11679)
- Jiang JJ, Li ZY, Liu XM (2022) Deep learning based monocular depth estimation: a survey. *Chin J Comput* 45(6):1276–1307. <https://doi.org/10.11897/SP.J.1016.2022.01276>
- Jiang YWQ, Tu JD, Liu Y, Gao XF, Long XX, Wang WP, Ma YX (2023) GaussianShader: 3D Gaussian splatting with shading functions for reflective surfaces. arXiv preprint [arXiv:2311.17977](https://arxiv.org/abs/2311.17977)
- Jiao JH, Wei HX, Hu TS, et al. (2022) FusionPortable: a multi-sensor campus-scene dataset for evaluation of localization and mapping accuracy on diverse platforms. In: IEEE/RSJ international conference on intelligent robots and systems (IROS), pp 4003–4009
- Jin WK, Zhao Z, Cao XC, Zhu JM, He XQ, Zhuang YT (2021) Adaptive spatio-temporal graph enhanced vision-language representation for video QA. *IEEE Trans Image Process* 30:5477–5489. <https://doi.org/10.1109/TIP.2021.3076556>
- Justin K, Chung MK, Ken G, Angjoo K, Matthew T (2023) LERF: Language embedded radiance fields. In: IEEE conference on computer vision and pattern recognition (CVPR), pp 19729–19739
- Keetha N, Karhade J, Murthy KJ, Yang GS, Scherer S, Ramanan D, Luiten J. (2023) SplaTAM: splat, track & map 3D Gaussians for dense RGB-D SLAM. arXiv preprint [arXiv:2312.02126](https://arxiv.org/abs/2312.02126)
- Kendall A, Martirosyan H, Dasgupta S, Henry P, Kennedy R, Bachrach A, Bry A (2017) End-to-end learning of geometry and context for deep stereo regression. In: IEEE international conference on computer vision (ICCV), pp 66–75
- Kerbl B, Kopanas G, Leimkuehler T, Drettakis G (2023) 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans Graph* 42(4):1–14. <https://doi.org/10.1145/3592433>
- Kim H, Lee D, Kang SY, Kim P (2023) Complex-motion NeRF: joint reconstruction and pose optimization with motion and depth priors. *IEEE Access* 11:97425–97434. <https://doi.org/10.1109/ACCESS.2023.3313184>
- Knapitsch A, Park J, Zhou QY, Koltun V (2017) Tanks and temples: benchmarking large-scale scene reconstruction. *ACM Trans Graph* 36(4):1–13. <https://doi.org/10.1145/3072959.3073599>
- Lee JC, Rho D, Sun XY, Hwan KJ, Park E (2023) Compact 3D Gaussian representation for radiance field. arXiv preprint [arXiv:2311.13681](https://arxiv.org/abs/2311.13681)
- Lee B, Lee H, Sun XY, Ali U, Park E (2024). Deblurring 3D Gaussian splatting. arXiv preprint [arXiv:2401.00834](https://arxiv.org/abs/2401.00834)
- Lei XH, Wang M, Zhou WG, Li HQ (2024) GaussNav: Gaussian splatting for visual navigation. arXiv preprint [arXiv:2403.11625v2](https://arxiv.org/abs/2403.11625v2)

- Li YM, Yang M, Zhang ZF (2019) A survey of multi-view representation learning. *IEEE Trans Knowl Data Eng* 31(10):1863–1883. <https://doi.org/10.1109/TKDE.2018.2872063>
- Li XC, Xiao YX, Wang BB, Ren HJ, Zhang YY, Ji JM (2023b) Automatic targetless LiDAR-camera calibration: a survey. *Artif Intell Rev* 56(9):9949–9987. <https://doi.org/10.1007/s10462-022-10317-y>
- Li ZQ, Wang WH, Li HY, Xie E, Sima CH, Lu T, Qiao Y, Dai JF (2023) BEVFormer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. In: *European conference on computer vision (ECCV)*, pp 1–18
- Liang ZH, Zhang Q, Feng Y, Shan Y, Jia K (2023) GS-IR: 3D Gaussian splatting for inverse rendering. *arXiv preprint arXiv:2311.16473*
- Lin JQ, Li ZH, Tang X, et al. (2024) VastGaussian: Vast 3D Gaussians for large scene reconstruction. *arXiv preprint arXiv:2402.17427*
- Liu KH, Zhan FN, Zhang JH, Xu MY, Yu YC, Saddik EA, Theobalt C, Xing E, Lu SJ (2023) 3d open-vocabulary segmentation with foundation models. *arXiv preprint arXiv:2305.14093*
- Liu Y, Chen LZ, Xie HN, Li J (2023) Research on lightweight 3D reconstruction techniques based on Gaussian splatting. In: *International conference on advances in artificial intelligence and applications (AAIA)*, pp 186–194
- Liu Y, Guan H, Luo CC, Fan L, Peng JR, Zhang ZX (2024) CityGaussian: real-time high-quality large-scale scene rendering with Gaussians. In: *European conference on computer vision (ECCV)*
- Lobay A, Forsyth DA (2006) Shape from texture without boundaries. *Int J Comput vis* 67(1):71–91. <https://doi.org/10.1007/s11263-006-4068-8>
- Long Q (1996) Conic reconstruction and correspondence from two views. *IEEE Trans Pattern Anal Mach Intell* 18(2):151–159. <https://doi.org/10.1109/34.481540>
- Ma XJ, Gong Y, Wang QR, Huang JW, Chen L, Yu F (2021) EPP-MVSNet: epipolar-assembling based depth prediction for multi-view stereo. In: *IEEE international conference on computer vision (ICCV)*, pp 5732–5740
- Ma L, Li XY, Liao J, Zhang Q, Wang X, Wang J, Sander PV (2022) Deblur-NeRF: neural radiance fields from blurry images. In: *IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 12861–12870
- Mai F, Hung YS, Chesi G (2010) Projective reconstruction of ellipses from multiple images. *Pattern Recogn* 43(3):545–556. <https://doi.org/10.1016/j.patcog.2009.07.003>
- Mevey ES, Jarvis GL (1997) Ranking of pattern for use in automation. *IEEE Trans. Ind. Electron. Control Instrum* 24(2):211–213. https://jglobal.jst.go.jp/en/detail?JGLOBAL_ID=201002022832460792
- Mildenhall B, Srinivasan PP, Tancik M, Barron JT, Ramamoorthi R, Ng R (2020) NeRF: representing scenes as neural radiance fields for view synthesis. In: *European conference on computer vision (ECCV)*, pp 405–421
- Mildenhall B, Srinivasan PP, Tancik M, Barron JT, Ramamoorthi R, Ng R, (2020) Nerf: representing scenes as neural radiance fields for view synthesis. In: *European conference on computer vision (ECCV)*, pp 99–106
- Mueller T, Evans A, Schied C, Keller A (2022) Instant neural graphics primitives with a multiresolution hash encoding. *ACM Trans Graph* 41(4):1–15. <https://doi.org/10.1145/3528223.3530127>
- Mueller T, Evans A, Schied C, Keller A (2023) Instant neural graphics primitives with a multiresolution hash encoding. *ACM Trans Graph* 41(4):1–15. <https://doi.org/10.1145/3528223.3530127>
- Mustafa A, Volino M, Kim H, Guillemaut JY, Hilton A (2021) Temporally coherent general dynamic scene reconstruction. *Int J Comput vis* 19(1):123–141. <https://doi.org/10.1007/s11263-020-01367-2>
- Nebauer C (1998) Evaluation of convolutional neural networks for visual recognition. *IEEE Trans Neural Netw* 9(4):685–696. <https://doi.org/10.1109/72.701181>
- Newcombe RA, Fox D, Seitz SM (2015) DynamicFusion: Reconstruction and tracking of non-rigid scenes in real-time. In: *IEEE conference on computer vision and pattern recognition (CVPR)*, pp 343–352
- Park K, Sinha U, Hedman P, Barron JT, Bouaziz S, Goldman DB, Martin BR, Seitz SM (2021) HyperNeRF: a higher-dimensional representation for topologically varying neural radiance fields. *ACM Trans Graph* 40(6):238–250. <https://doi.org/10.1145/3478513.3480487>
- Phothong W, Wu TC, Yu CY, Lai JY, Wang DW, Liao CY (2018) Generation and quality improvement of 3D models from silhouettes of 2D images. *J Chin Inst Eng* 41(3):216–228. <https://doi.org/10.1080/02533839.2018.1459848>
- Quentin H, Moussab B, Arthur M, et al. (2024) 3DGS-Calib: 3D Gaussian splatting for multimodal spatio-temporal calibration. *arXiv preprint arXiv:2403.11577*
- Reizenstein J, Shapovalov R, Henzler P, Sbordon L, Labatut P, Novotny D (2021) Common objects in 3D: large-scale learning and evaluation of real-life 3D category reconstruction. In: *IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 10901–10911
- Ren KR, Jiang LH, Lu T, Yu ML, Xu LN, Ni ZK, Dai B. (2024) Octree-GS: towards consistent real-time rendering with LOD-structured 3D Gaussians. *arXiv preprint arXiv:2403.17898*

- Sebastian K, Narunas V, Mirco C, Pedro H, Timo R (2024) Open3DSG: open-vocabulary 3D scene graphs from point clouds with queryable objects and open-set relationships. arXiv preprint [arXiv:2402.12259](https://arxiv.org/abs/2402.12259)
- Seongbo H, Jiung Y, Hyeonwoo Y (2024) RGBD GS-ICP SLAM. arXiv preprint [arXiv:2403.12550](https://arxiv.org/abs/2403.12550)
- Shen QH, Wu ZK, Yi XY, Zhou P, Zhang HW, Yan SC, Wang XC (2024) Gamba: marry Gaussian splatting with mamba for single view 3D reconstruction. arXiv preprint [arXiv:2403.18795](https://arxiv.org/abs/2403.18795)
- Shi JC, Wang M, Duan HB, Guan SH (2023) Language embedded 3D Gaussians for open-vocabulary scene understanding. arXiv preprint [arXiv: 2311.18482](https://arxiv.org/abs/2311.18482)
- Shirpour M, Khairdoost N, Bauer MA, Beauchemin SS (2023) Traffic object detection and recognition based on the attentional visual field of drivers. *IEEE Trans Intell Veh* 8(1):594–604. <https://doi.org/10.1109/TIV.2021.3133849>
- Somaya AM, Rayana B, Suchithra K, Ahmed B (2018) Robust feature point detectors for car make recognition. *Comput Ind* 100:129–136. <https://doi.org/10.1016/j.compind.2018.04.014>
- Song HS, Wang X, Hua C, Wang WX, Guan Q, Zhang ZY (2018) Vehicle trajectory clustering based on 3D information via a coarse-to-fine strategy. *Soft Comput* 22(5):1433–1444. <https://doi.org/10.1007/s00500-017-2831-0>
- Stanislaw S, Christian R, Andrea V (2023). Splatter image: ultra-fast single-view 3D reconstruction. arXiv preprint [arXiv:2312.13150](https://arxiv.org/abs/2312.13150)
- Straub J, Whelan T, Ma LN, et al. (2019) The replica dataset: a digital replica of indoor spaces. arXiv preprint [arXiv:1906.05797](https://arxiv.org/abs/1906.05797)
- Sturm J, Engelhard N, Endres F, Burgard W, Cremers D (2012) A benchmark for the evaluation of RGB-D SLAM systems. In: *IEEE/RSJ international conference on intelligent robots and systems (IROS)*, pp 573–580
- Suhail M, Esteves C, Sigal L, Makadia A (2022) Generalizable patch-based neural rendering. In: *European conference on computer vision (ECCV)*, pp 156–174
- Tang JX, Ren JW, Zhou H, Liu ZW, Zeng G (2023) DreamGaussian: generative Gaussian splatting for efficient 3D content creation. arXiv preprint [arXiv:2309.16653](https://arxiv.org/abs/2309.16653)
- Teoh ER (2020) What's in a name? Drivers' perceptions of the use of five SAE Level 2 driving automation systems. *J Saf Res* 72:14–151. <https://doi.org/10.1016/j.jsr.2019.11.005>
- Tosi F, Zhang YM, Gong ZR, Sandström E, Mattoccia S, Oswald MR, Poggi R (2024) How NeRFs and 3D Gaussian splatting are reshaping SLAM: a survey. arXiv preprint [arXiv:2402.13255](https://arxiv.org/abs/2402.13255)
- Trabelsi R, Khemmar R, Decoux B, Ertaud JY, Butteau R (2022) Recent advances in vision-based On-road behaviors understanding: a critical survey. *Sensors* 22(7):2654. <https://doi.org/10.3390/s22072654>
- Vladimir Y, Yue L, Theo G, Martin R (2023) Oswald. Gaussian-SLAM: photo-realistic dense SLAM with Gaussian splatting. arXiv preprint [arXiv:2312.10070](https://arxiv.org/abs/2312.10070)
- Wang YF, Serena F, Wu SH, Oeztirceli C, Sorkine HO (2019) Differentiable surface splatting for point-based geometry processing. *ACM Trans Graph* 38(6):1–14. <https://doi.org/10.1145/3355089.3356513>
- Wang X, Wang C, Liu B, Zhou XQ, Zhang L, Zheng J, Bai X (2021a) Multi-view stereo in the deep learning era: a comprehensive review. *Displays* 70:243188826. <https://doi.org/10.1016/j.displa.2021.102102>
- Wang ZG, Zhan J, Duan CG, Guan X, Yang K (2022) Vehicle detection in severe weather based on pseudo-visual search and HOG-LBP feature fusion. *P i Mech Eng D-J Aut* 236(7):1607–1618. <https://doi.org/10.1177/09544070211036311>
- Wang H, Ji HB, Gao XB (2005) A novel splatting algorithm based on directional reconstruction filter kernels. *J Xidian Univer* 32(4):532–537. <https://journal.xidian.edu.cn/xdxb/CN/Y2005/V32/I4/532>
- Wang QQ, Wang ZC, Genova K, et al. (2021) IBRNet: learning multi-view image-based rendering. In: *IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 4690–4699
- Wang HY, Wang JW, Agapito L (2023) Co-SLAM: joint coordinate and sparse parametric encodings for neural real-time SLAM. In: *IEEE conference on computer vision and pattern recognition (CVPR)*, pp 13293–13302
- Wang YX, Yi XY, Wu ZK, Zhao N, Chen L, Zhang HW (2024) View-consistent 3D editing with Gaussian splatting. arXiv preprint [arXiv:2403.11868](https://arxiv.org/abs/2403.11868)
- Wu GC, Yi TR, Fang JM, Xie LX, Zhang XP, Wei W, Liu WY, Tian Q, Wang XG (2023) 4D Gaussian splatting for real-time dynamic scene rendering. arXiv preprint [arXiv:2310.08528](https://arxiv.org/abs/2310.08528)
- Wu T, Yuan YJ, Zhang LX, Yang J, Cao YP, Yan LQ, Gao L (2024) Recent advances in 3D Gaussian splatting. arXiv preprint [arXiv:2403.11134](https://arxiv.org/abs/2403.11134)
- Wu J, Bian JW, Li XH, Wang GR, Reid I, Torr P, Victor AP (2024) GaussCtrl: multi-view consistent text-driven 3D Gaussian splatting editing. arXiv preprint [arXiv:2403.08733](https://arxiv.org/abs/2403.08733)
- Xie TY, Zong ZS, Qiu YX, Li X, Feng YT, Yang Y, Jiang CFF (2023) PhysGaussian: physics-integrated 3D Gaussians for generative dynamics. arXiv preprint [arXiv:2311.12198](https://arxiv.org/abs/2311.12198)
- Yan XQ, Hu SZ, Mao YQ, Ye YD, Yu H (2021) Deep multi-view learning methods: a review. *Neurocomputing* 448:106–129. <https://doi.org/10.1016/j.neucom.2021.03.090>

- Yan L, Hu X, Zhao LY, Chen Y, Wei PC, Xie H (2022) DGS-SLAM: a fast and robust RGBD SLAM in dynamic environments combined by geometric and semantic information. *Remote Sens* 14(3):795. <https://doi.org/10.3390/rs1403079>
- Yan C, Qu DL, Xu D, Zhao B, Wang ZG, Wang D, Li XL (2023) GS-SLAM: dense visual SLAM with 3D Gaussian splatting. arXiv preprint [arXiv:2311.11700](https://arxiv.org/abs/2311.11700)
- Yan YZ, Lin HT, Zhou CX, Wang WJ, Sun HY, Zhan K, Lang XP, Zhou XW, Peng SD (2024). Street Gaussians for modeling dynamic urban scenes. arXiv preprint [arXiv:2401.01339](https://arxiv.org/abs/2401.01339)
- Yang GW, Zhou WY, Peng HY, Liang D, Mu TJ, Hu SM (2023a) Recursive-NeRF: an efficient and dynamically growing NeRF. *IEEE Trans vis Comput Graph* 29(12):5124–5136. <https://doi.org/10.1109/TVCG.2022.3204608>
- Yang XR, Li H, Zhai HJ, Ming YH, Liu YQ, Zhang GF (2022) Vox-Fusion: dense tracking and mapping with voxel-based neural implicit representation. In: *IEEE international symposium on mixed and augmented reality (ISMAR)*, pp 499–507
- Yang ZY, Gao XY, Zhou W, Jiao SH, Zhang YQ, Jin XG (2023). Deformable 3D Gaussians for high-fidelity monocular dynamic scene reconstruction. arXiv preprint [arXiv:2309.13101](https://arxiv.org/abs/2309.13101)
- Yang H, Hong LQ, Li AX, Hu TY, Li ZG, Lee GH, Wang LW (2023) ContraNeRF: generalizable neural radiance fields for synthetic-to-real novel view synthesis via contrastive learning. In: *IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 16508–16517
- Yang ZY, Gao XY, Sun YT, et al. (2023) Spec-Gaussian: Anisotropic view-dependent appearance for 3D Gaussian splatting. arXiv preprint [arXiv:2311.16473](https://arxiv.org/abs/2311.16473)
- Yang ZY, Yang HY, Pan ZJ, Zhang L (2023) Real-time photorealistic dynamic scene representation and rendering with 4D Gaussian splatting. arXiv preprint [arXiv:2310.10642](https://arxiv.org/abs/2310.10642)
- Yao Y, Luo ZX, Li SW, Fang T, Quan L (2018) MVSNet: depth inference for unstructured multi-view stereo. In: *15th European conference on computer vision (ECCV)*, pp 785–801
- Yao Y, Luo ZX, Li SW, Fang T, Quan L (2019) Recurrent MVSNet for high-resolution multi-view stereo depth inference. In: *IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 5525–5534
- Ye MQ, Danelljan M, Yu F, Ke L (2023) Gaussian grouping: segment and edit anything in 3D scenes. arXiv preprint [arXiv:2312.00732](https://arxiv.org/abs/2312.00732)
- Yeshwanth C, Liu YC, Niesner M, Dai A (2023) Scannet++: a high-fidelity dataset of 3d indoor scenes. In: *IEEE/CVF international conference on computer vision (ICCV)*, pp 12–22
- Young RA (1986) Locating industrial parts with subpixel accuracies. *Proc SPIE—Int Soc Opt Eng* 728:2–9. <https://doi.org/10.1117/12.937817>
- Yu ZH, Gao SH (2020) Fast-MVSNet: sparse-to-dense multi-view stereo with learned propagation and Gauss-Newton refinement. In: *IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pp 1949–1958
- Yu ZH, Chen AP, Huang BB, Sattler T, Geiger A (2023). Mip-splatting: alias-free 3D Gaussian splatting. arXiv preprint [arXiv:2311.16493](https://arxiv.org/abs/2311.16493)
- Yu ML, Lu T, Xu LN, Jiang LN, Xiangli YB, Dai B. (2024) GSDF: 3DGS meets SDF for improved rendering and reconstruction. arXiv preprint [arXiv:2403.16964](https://arxiv.org/abs/2403.16964)
- Zhai GD, Zhang WT, Hu WY, Ji ZD (2020) Coal mine rescue robots based on binocular vision: a review of the state of the art. *IEEE Access* 8:130561–130575. <https://doi.org/10.1109/ACCESS.2020.3009387>
- Zhang H, Li F, Zhao JH, Shen DM, Liu YB, Yu T (2023b) Controllable free viewpoint video reconstruction based on neural radiance fields and motion graphs. *IEEE Trans vis Comput Graph* 29(12):4891–4905. <https://doi.org/10.1109/TVCG.2022.3192713>
- Zhang H, Li F, Zhao JH, Shen DM, Liu YB, Yu T (2023c) Controllable free viewpoint video reconstruction based on neural radiance fields and motion graphs. *IEEE Trans vis Comput Graph* 29(12):4891–4905. <https://doi.org/10.1109/TVCG.2022.3192713>
- Zhang MT, Zheng SH, Bao ZP, Hebert M, Wang YX (2023) Beyond RGB: scene-property synthesis with neural radiance fields. In: *23rd IEEE/CVF winter conference on applications of computer vision (WACV)*, pp 795–805
- Zhang ZW, Zhang ZZ, Yu Q, Yi R, Xie Y, Ma LZ (2023) LiDAR camera panoptic segmentation via geometry-consistent and semantic aware alignment. In: *international conference on computer vision (ICCV)*, pp 3662–3671
- Zhao LZ, Wang P, Liu PD (2024) BAD-Gaussians: bundle adjusted deblur Gaussian splatting. arXiv preprint [arXiv:2403.11831](https://arxiv.org/abs/2403.11831)
- Zheng TX, Huang S, Li YF, Feng MC, (2020) Key techniques for vision based 3D reconstruction: a review. *Acta Anat. Sin* 46(4):631–652. <https://doi.org/10.16383/j.aas.2017.c170502>
- Zheng CR, Zhu QY, Xu W, Liu XY, Guo QZ, Zhang F (2022) FAST-LIVO: fast and tightly-coupled sparse-direct LiDAR-inertial-visual odometry. In: *IEEE/RSJ international conference on intelligent robots and systems (IROS)*, pp 4003–4009

- Zhou LL, Wu GX, Zuo YB, Chen XY, Hu HL (2024a) A comprehensive review of vision-based 3d reconstruction methods. *Sensors-Basel* 24(7):2314. <https://doi.org/10.3390/s24072314>
- Zhou XY, Lin ZW, Shan XJ, Wang YT, Sun DQ, Yang MH (2023) DrivingGaussian: composite Gaussian splatting for surrounding dynamic autonomous driving scenes. arXiv preprint [arXiv:2312.07920](https://arxiv.org/abs/2312.07920)
- Zhou HY, Shao JH, Xu L, et al. (2024b) HUGS: holistic urban 3D scene understanding via Gaussian splatting. arXiv preprint [arXiv:2403.12722](https://arxiv.org/abs/2403.12722)
- Zhu F, Guo S, Song L, Xu K, Hu JY (2023a) Deep review and analysis of recent NeRFs. *APSIPA Trans Signal Inf Process* 12(1):1–32. <https://doi.org/10.1561/116.00000162>
- Zhu ZH, Peng SY, Larsson V, Xu WW, Bao HJ, Cui ZP, Oswald MR, Pollefeys M (2022) NICE-SLAM: neural implicit scalable encoding for SLAM. In: *IEEE conference on computer vision and pattern recognition (CVPR)*, pp 12786–12796
- Zhu ZY, Ma XJ, Chen YX, Deng ZD, Huang SY, Li, Qing (2023) 3D-VisTA: pre-trained transformer for 3D vision and text alignment. In: *International conference on computer vision (ICCV)*, pp 2911–2921
- Zou ZX, Yu ZP, Guo YC, Li YG, Liang D, Cao YP, Zhang SH (2023) Triplane meets Gaussian splatting: fast and generalizable single-view 3D reconstruction with transformers. arXiv preprint [arXiv:2312.09147](https://arxiv.org/abs/2312.09147)
- Zwicker M, Pfister H, Baar VJ, Gross M (2002a) EWA splatting. *IEEE Trans vis Comput Graph* 8(3):223–238. <https://doi.org/10.1109/TVCG.2002.1021576>
- Zwicker M, Pfister H, Baar VJ, Gross M (2002b) EWA splatting. *IEEE Trans vis Comput Graph* 8(3):223–238. <https://doi.org/10.1109/TVCG.2002.1021576>
- Zwicker M, Pfister H, Baar VJ, Gross M (2001) EWA volume splatting. In: *IEEE visualization conference (VIS)*, pp 29–36

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.